The spectral measure of an elliptic orbital integral, via Arthur's local trace formula*

Draft 4

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Abstract

This paper was written with the help of AI systems: Claude (Anthropic) suggested the approach and drafted the paper as a co-author, and Gemini (Google) refereed the third draft. See the Acknowledgments for details. The paper revises and updates the first author's earlier paper [Jo92], using a method different from the one used in the analogous paper [JC26]. Let G be an unramified semisimple group over a p-adic field, K a hyperspecial maximal compact subgroup, and γ a regular elliptic element. Taking one test function spherical and the other supported on the elliptic set, we specialize Arthur's local trace formula. On the geometric side only the orbital integrals of G survive. On the spectral side only two kinds of terms survive: the unramified principal series of G, weighted by a weighted character of the elliptic test function, and Arthur's elliptic tempered virtual characters with K-fixed vectors. We deduce that the spherical orbital integrals of γ are given by a finite measure on the unitary unramified characters. In rank one, that is for $SL(2)$ and $PGL(2)$, its absolutely continuous part has density a normalized weighted character. In higher rank the density comes from Arthur's unnormalized weighted character, and its relation to the normalized one has not yet been computed. Its singular part is supported on subtori where the R-groups of the unramified principal series are non-trivial, and it vanishes when the center of G is connected. For $SL(n)$ the singular part is given by κ-orbital integrals. In that case we determine the elliptic data explicitly, and we show that they have the same index set and the same supports as the κ-orbital integrals. We compare the result with the truncation method of the companion paper [JC26], and work out $PGL(2)$, $SL(2)$, $PGL(3)$ and $SL(3)$.

This is a working draft. The results are proved modulo normalizations in Arthur's formula that have not yet been checked. The main theorem is proved in outline. For $SL(n)$ the identification of the densities is given in outline only. The section "Status and what remains" lists what is missing.

*This is a working draft. Statements are labeled as proved here, quoted (with a reference), or expected; every "Status" paragraph says which, and what remains to be checked.

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‡Claude is an AI model made by Anthropic. See the Acknowledgments for the division of work.

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1 Introduction

This introduction states the problem and recalls the truncation method of the companion paper. It then describes the method of this paper, states the results, and gives a plan of the paper.

The problem. Let F be a p-adic field with ring of integers $\mathcal{O}$ and residue field of order q. Let G be a connected unramified semisimple group over F, and $K \subset G(F)$ a hyperspecial maximal compact subgroup. Let $\widehat{T}^u$ be the compact torus of unitary unramified characters of a minimal Levi subgroup M_0, with Weyl group W. For f in the spherical Hecke algebra $\mathscr{H}(G,K)$ let $f^\vee$ be its Satake transform, a W-invariant function on $\widehat{T}^u$, and let $d\mu$ be the Plancherel measure on $\widehat{T}^u/W$. For $\gamma \in G(F)$ regular elliptic, the orbital integral $f \mapsto O_\gamma(f) = \int_{G(F)} f(g^{-1}\gamma g)\,dg$ is a linear form on $\mathscr{H}(G,K)$, i.e. on W-invariant Laurent polynomials on $\widehat{T}^u$. We ask what kind of object it is:

$$O_\gamma(f) = \int_{\widehat{T}^u/W} f^\vee(\chi)\,dI_\gamma(\chi) \qquad (f \in \mathscr{H}(G,K)), \tag{1.1}$$

with dI_γ a measure, and if so which measure.

Route A. The companion paper [JC26] (a revision of [Jo92]) attacks (1.1) directly. It truncates the integral over g to a finite union $G(T)$ of double cosets, and expands the spherical function in matrix coefficients. It then uses Casselman's asymptotics and Brion's theorem to write the truncated kernel $I^T(\chi,\gamma)$ as an exponential polynomial in T. The T-independent part $I(\chi,\gamma)$ is an explicit rational function of χ. Under a regularity hypothesis (R) on the oscillating terms, [JC26, Thm. 4.5] proves $dI_\gamma = I(\chi,\gamma)\,d\mu$. The hypothesis holds for $PGL(2)$. It fails for $SL(2)$, where dI_γ has an atom at the unramified quadratic character. It also fails for $PGL(3)$, where nevertheless dI_γ is absolutely continuous [JC26, App. A]. So (R) is not the right condition, and [JC26, Conj. 4.6] predicts the general shape of dI_γ.

Route B. Here we derive (1.1) from Arthur's local trace formula [Ar89a, Ar91], in the form given by his theory of elliptic tempered characters [Ar93]. Arthur's formula is an identity between a geometric and a spectral expression, both of them distributions on $G(F)\times G(F)$. For $f = f_1 \otimes f_2$ it reads, schematically,

$$\sum_M c_M \int_{\Gamma_{\mathrm{ell}}(M)} J_M(\gamma,f)\,d\gamma = \sum_M c_M \int_{\Pi(M)} J_M(\pi,f)\,d\pi,$$

with weighted orbital integrals on the left and weighted characters on the right. Take $f_1 = f$ spherical and $f_2 = \phi$ supported on the regular elliptic set. Then almost everything vanishes.

- On the geometric side only $M = G$ survives, and it is the elliptic orbital integral $I(f,\phi)$ (§5).

- On the spectral side, a weighted character of the spherical function f vanishes unless its weight is trivial (Lemma 3.4). This rests on the fact that the canonically normalized unramified intertwining operators fix the spherical vector. A character of ϕ vanishes on properly induced representations. What remains is a sum over the unitary unramified characters χ and over the finitely many elliptic tempered virtual characters with K-fixed vectors (§6).

Results. The statements below are proved here modulo the normalizations listed in §12, which are being checked. In rank ≥ 2 the formula of Theorem B is also subject to the correction discussed in Remark 4.1.

Theorem A (existence)**.** *For every $\phi \in C_c^\infty(G_{\mathrm{ell}})$ the linear form $f^\vee \mapsto I(f,\phi)$ is given by a finite complex measure dI_ϕ on $\widehat{T}^u/W$.*

Theorem B (the absolutely continuous part)**.** $dI_\phi = I_B(\chi,\phi)\,d\mu(\chi) + dI_\phi^{\mathrm{sing}}$*, where*

$$I_B(\chi,\phi)\,\rho(\chi) = c_G\, J_{M_0}(\chi^{-1},\phi), \qquad c_G = (-1)^{\dim \mathfrak{a}_0^G}|W|^{-1},$$

ρ is the Plancherel density, and J_{M_0} is Arthur's weighted character for the canonically normalized unramified intertwining operators (§3.3). In particular $I_B\rho$ is a smooth function on $\widehat{T}^u$. The formula for $I_B\rho$ is asserted in rank one. In rank ≥ 2 the function $J_{M_0}(\chi^{-1},\phi)$ has to be replaced by the one that comes from Arthur's unnormalized weighted character (Remark 4.1).

Theorem C (the singular part)**.** dI_ϕ^{sing} *is a finite sum of smooth measures on subtori of $\widehat{T}^u/W$, one for each elliptic tempered virtual character, with K-fixed vectors, of a Levi subgroup $M \supsetneq M_0$ (§6.3). The subtorus attached to M has dimension* $\dim \mathfrak{a}_M^G$*. If all R-groups of the unitary unramified principal series are trivial, for instance if $Z(G)$ is connected, then $dI_\phi^{\mathrm{sing}} = 0$.*

Theorem D ($SL(n)$)**.** *For $G = SL(n)$, $dI_\gamma^{\mathrm{sing}}$ is the sum over the non-trivial unramified κ of the push-forwards of the stable spectral measures of the elliptic endoscopic groups, as in [JC26, Conj. 4.6(c)].*

The comparison with Route A is Conjecture 8.1: the function I_B of Theorem B equals the T-independent part I of the truncated kernel of [JC26]. If it holds, Route A gives closed formulas for the density, and Route B explains why it is smooth and what replaces it where (R) fails. Proposition 8.3 shows that the two densities agree whenever both routes are known to produce a measure, and Corollary 8.5 deduces the conjecture for $PGL(2)$ from Theorem 7.1. For $SL(n)$, Proposition 9.6 matches the index sets and the supports of the two descriptions of the singular part, and Proposition 9.7 reduces Theorem D to Theorem 7.1 and to statements of Route A.

Scope: rank one and higher rank. Some statements of this paper are complete only in rank one, that is, for $SL(2)$ and $PGL(2)$. The division is as follows.

- *All ranks.* The specialization of the geometric side (Proposition 5.1), the existence of the spectral measure (Theorem A), the smoothness of the density, and the supports of the singular part (Theorem C) are stated for all G, modulo the items of §12. So are the R-groups, the elliptic data and the dictionary for $SL(n)$ (§§9.1–9.3), for every n.

- *Rank one only.* The formula of Theorem B for the density as a normalized weighted character is established, modulo the same items, only in rank one. Arthur's formula uses unnormalized weighted characters. In rank one the difference does not matter for our test functions. In rank ≥ 2 it produces further terms that have not been computed (Remark 4.1). The first term of Theorem 7.1, the displayed form of Conjecture 8.1, and the statements about the density in §§10.3–10.4 are therefore provisional in rank ≥ 2.
- *$PGL(2)$ and $SL(2)$.* Corollary 8.5, which deduces Conjecture 8.1 from Theorem 7.1, is for $PGL(2)$. The examples that are worked out completely are §§10.1–10.2. All examples in §§3–8 are in rank one.
- *$SL(n)$ with n prime.* The explicit form (9.2) of Theorem D is for prime n, and its normalizations have been checked only for $n = 2$.

Why this is useful. Arthur's local trace formula is one of the deepest tools of local harmonic analysis, and most of its terms are hard to compute. In the case treated here every surviving term can be made explicit. The paper is meant both as an independent proof of (1.1) and as an introduction to Arthur's local theory through a concrete case.

Plan. §2 reviews the literature before and after 1997. §3 collects the unramified background: normalized intertwining operators, R-groups, and weighted characters. §4 states Arthur's local trace formula. §5 and §6 specialize its two sides. §7 assembles Theorems A–C. §8 compares the result with Route A. §9 treats $SL(n)$. §10 works out examples. §11 summarizes the results and their status. §12 lists what remains to be checked, and Appendix A is a glossary.

2 Literature review

The first version of the companion paper was published in 1992 [Jo92] and corrected in 1997. Arthur's local trace formula and his theory of elliptic tempered characters appeared in between. So we divide the literature at 1997. §2.1 describes the tools available when the earlier versions were written, and §2.2 describes what came later and bears on Route B.

2.1 Before 1997

This part describes the tools that were available when the earlier versions of the companion paper were written. They are the spherical theory, intertwining operators and R-groups, Arthur's trace formula, and elliptic harmonic analysis.

Spherical harmonic analysis. Satake identified the spherical Hecke algebra with W-invariant functions on the torus of unramified characters [Sa63]. Macdonald gave the explicit formulas for the spherical function and the spherical Plancherel measure [Ma71]. Casselman re-derived Macdonald's formula from intertwining operators and the asymptotics of matrix coefficients [Ca80], using the theory of Jacquet modules in his notes [Ca95]. Cartier's survey [Ca79] collects the unramified theory. Bernstein and Zelevinsky's geometric lemma [BZ77] describes Jacquet modules of induced representations.

Intertwining operators and R-groups. Knapp and Stein introduced the R-group governing the reducibility of induced representations for real groups [KS71], and Silberger proved the p-adic analogue of their dimension theorem [Si78]. Keys computed the R-groups of the unramified principal series of Chevalley groups [Ke82]. Langlands' normalizing factors for intertwining operators, built from L-functions, go back to his work on Eisenstein series [La76]. Van Dijk's formula for induced characters [vD72] gives the vanishing of characters of cuspidal functions on properly induced representations.

Arthur's trace formula and (G,M)-families. The invariant trace formula [Ar81, Ar88] introduced (G,M)-families, weighted orbital integrals and weighted characters, and the splitting formulas. Arthur studied weighted characters as functions of the representation in [Ar89b]. The local trace formula was announced in [Ar89a] and proved in [Ar91]. Arthur then reorganized its spectral side in terms of elliptic tempered virtual characters [Ar93], and studied the Fourier transforms of weighted orbital integrals [Ar94]. His paper on local character relations [Ar96] relates these objects to endoscopic transfer.

Elliptic harmonic analysis and endoscopy. Kazhdan's cuspidal geometry [Ka86] and the trace Paley–Wiener theorem of Bernstein, Deligne and Kazhdan [BDK86] control the elliptic part of harmonic analysis. Labesse and Langlands worked out L-indistinguishability and κ-orbital integrals for $SL(2)$ [LL79], and Langlands and Shelstad defined transfer factors [LS87]. Waldspurger computed the spherical orbital integrals of $GL(N)$ spectrally [Wa89a], by the truncation method that Route A follows.

2.2 After 1997

This part describes later work that bears on Route B. It covers normalizations and transfer, the fundamental lemma, elliptic representations, the Plancherel formula, and variants of the local trace formula.

Normalization and transfer of weighted characters. Arthur proposed a canonical normalization of weighted characters, and a conjectural transfer identity for them [Ar98]. He then studied the transfer of weighted orbital integrals [Ar99], and stabilized the trace formula in a series of three papers [Ar02, Ar01, Ar03]. The stabilization was completed for twisted groups by Mœglin and Waldspurger [MW16]. Arthur's classification of representations of classical groups [Ar13] uses the stabilized local and global trace formulas.

The fundamental lemma. Ngô's proof of the fundamental lemma [Ng10], combined with earlier reductions (see [JC26, Literature review]), made the endoscopic statements of §9 unconditional at the level of orbital integrals.

Elliptic representations. Reeder related the Euler–Poincaré pairing of Iwahori-spherical representations to elliptic representations of Weyl groups [Re01]. Opdam and Solleveld computed Ext groups of tempered representations of affine Hecke algebras, recovering Arthur's elliptic pairing

[OS13]. These results describe exactly the elliptic virtual characters with Iwahori-fixed vectors, a class containing those that appear in Theorem C.

The Plancherel formula. Waldspurger's exposition of Harish-Chandra's Plancherel formula [Wa03] supplies complete proofs of the harmonic analysis behind Arthur's spectral side.

Variants of the local trace formula. Waldspurger proved a local trace formula adapted to the Gross–Prasad conjecture [Wa10]. Beuzart-Plessis proved one for the unitary Gan–Gross–Prasad conjecture [BP20]. Peng made the spectral side of the stable local trace formula explicit when one test function is cuspidal [Pe16]. H. Li proved a local trace formula for infinitesimal symmetric spaces [Li21].

Route A. The companion paper [JC26] revises [Jo92] by the truncation method, with the regularity hypothesis (R), and with examples in ranks one and two.

Status 2.1. The entries cited in this section have not yet all been checked against publisher records. The page ranges of [Ar01, Ar02, Ar03], [MW16] and [Wa10] in particular are to be verified.

3 Unramified background

This section collects the facts about unramified principal series that the specialization in §§5–6 needs. The essential point is Lemma 3.4: with the canonical unramified normalization, a weighted character of a spherical function has no weight.

3.1 Notation

This subsection fixes the notation for the group, its parabolic subgroups and the unramified principal series. Example 3.1 makes the notation explicit in rank one, and Table 1 at the end of the subsection collects the symbols used in the paper.

We use the notation of [JC26, §2]. A is a maximal split torus, $M_0 = Z_G(A)$, $B = M_0N_0$ a Borel subgroup, W the Weyl group, $\mathfrak{a}_0 = X_*(A) \otimes \mathbb{R}$, and $\mathscr{P}(M)$ the set of parabolic subgroups with Levi component M. For a character χ of $M_0(F)$ and $P \in \mathscr{P}(M_0)$, $I_P(\chi)$ is the normalized induced representation, realized on functions on K (the compact picture). Φ_K is the spherical vector, equal to 1 on K. For unramified χ write $\chi_\alpha = \chi \circ \alpha^\vee$, an unramified character of $F^\times$, and $L(s,\chi_\alpha) = (1 - \chi_\alpha(\pi)q^{-s})^{-1}$. To keep formulas short we assume that G is split. The quasi-split unramified case changes only the local factors attached to roots.

Example 3.1 (rank one)**.** Let $G = SL(2)$ or $PGL(2)$, with A the diagonal torus and B the upper triangular Borel subgroup. Then $M_0 = A$, and $\mathscr{P}(A) = \{B, \bar{B}\}$ with $\bar{B}$ the lower triangular Borel subgroup. The Weyl group is $W = \{1, w\}$, the space $\mathfrak{a}_0$ is a line, and there is one reduced positive root α.

For $G = SL(2)$ an unramified character of $A(F)$ has the form $\chi(\mathrm{diag}(t,t^{-1})) = \mu(t)$, with μ an unramified character of $F^\times$. Put $z = \mu(\pi)$, where π is a uniformizer of F. Here $\alpha^\vee(t) = \mathrm{diag}(t,t^{-1})$, so $\chi_\alpha = \mu$ and $\chi_\alpha(\pi) = z$.

For $G = PGL(2)$ an unramified character of $A(F)$ has the form $\chi(\mathrm{diag}(a,b)) = \mu(a/b)$, and again we put $z = \mu(\pi)$. Here $\alpha^\vee(t)$ is the image of $\mathrm{diag}(t,t^{-1})$, so $\chi_\alpha = \mu^2$ and $\chi_\alpha(\pi) = z^2$.

In both cases χ is unitary exactly when $|z| = 1$, so $\widehat{T}^u$ is the unit circle. The element w sends z to z^{-1}, and a Satake transform $f^\vee$ is a Laurent polynomial in z that is invariant under $z \mapsto z^{-1}$. These are the coordinates of [JC26, §6.0].

Table 1: Notation. Symbols taken from the companion paper are defined there, at the place cited.

Symbol	Meaning
F, $\mathscr{O}$, q	A p-adic field, its ring of integers, and the order of its residue field (§1).
π	A uniformizer of F when it is the argument of a character, as in $\chi_\alpha(\pi)$. Elsewhere π, σ and τ denote representations or virtual characters.
G, K	A connected unramified semisimple group over F, and a hyperspecial maximal compact subgroup of $G(F)$.
A, M_0, B, W	A maximal split torus, its centralizer, a Borel subgroup $B = M_0N_0$, and the Weyl group.
$\mathscr{L}$, $\mathscr{L}(M)$, $\mathscr{P}(M)$, $\mathscr{P}^L(M)$	The Levi subgroups containing M_0. Those containing M. The parabolic subgroups with Levi component M. The same inside a Levi subgroup L.
$\mathfrak{a}_M$, $\mathfrak{a}_M^L$, $\mathfrak{a}_0$	$\mathfrak{a}_M = X_*(A_M) \otimes \mathbb{R}$, with A_M the split part of the center of M. $\mathfrak{a}_M^L$ is the kernel of $\mathfrak{a}_M \to \mathfrak{a}_L$. $\mathfrak{a}_0 = \mathfrak{a}_{M_0}$. Since G is semisimple, $\mathfrak{a}_0^G = \mathfrak{a}_0$.
H_M, H_P	The map $M(F) \to \mathfrak{a}_M$ with $e^{\langle H_M(m),\lambda\rangle} = \lvert\lambda(m)\rvert$ for rational characters λ of M, and its extension to $G(F)$ by $H_P(mnk) = H_M(m)$ [JC26, Glossary].
$\widehat{T}^u$, χ_Λ, χ_α	The compact torus of unitary unramified characters of $M_0(F)$. The twist $\chi \cdot e^{\Lambda(H_{M_0}(\cdot))}$. The character $\chi \circ \alpha^\vee$ of $F^\times$.
$\mathscr{H}(G,K)$, $f^\vee$	The spherical Hecke algebra and the Satake transform.
$d\mu$, ρ, $d\chi$	The Plancherel measure on $\widehat{T}^u/W$, its density, and Haar measure on $\widehat{T}^u$.
$I_P(\chi)$, Φ_K	The normalized induced representation in the compact picture, and its spherical vector.
$J_{P'\|P}$, $r_{P'\|P}$, $R_{P'\|P}$, c_α	The standard intertwining operator, the normalizing factor, the normalized operator, and the rank-one c-function (§3.2).
$\theta_Q(\Lambda)$	The product of the $\Lambda(\alpha^\vee)$ over the simple roots α of Q, divided by a covolume [Ar81, §6], [JC26, (2.16)].
$\mathscr{R}_M(\chi,P)$, $J_M(\chi,f)$, J_M^L	The limit of the (G,M)-family of normalized operators, the weighted character, and its analogue for a Levi subgroup $L \supseteq M$ (§3.3).
G_{ell}, O_γ, $I(f,\phi)$	The regular elliptic set, the orbital integral, and the distribution of Proposition 5.1.
$D(\gamma)$	The Weyl discriminant $\det(1 - \mathrm{Ad}(\gamma))$ on $\mathfrak{g}/\mathfrak{g}_\gamma$.
K_f, $u(x,T)$, $J^T(f)$, $J(f)$	The kernel, Arthur's truncation function, the truncated integral and its constant term (§4.1).
$J_M(\gamma,f)$, v_M, $\Gamma_{\mathrm{ell}}(M)$	The weighted orbital integral, its weight, and the elliptic classes of $M(F)$ (§4.2).
$J_{\mathrm{geom}}(f)$, $J_{\mathrm{spec}}(f)$	The two sides of Arthur's local trace formula, (4.1) and (4.2).
$\Pi_{\mathrm{disc}}(M)$, $a^M_{\mathrm{disc}}(\pi)$	Arthur's set of representations of $M(F) \times M(F)$ on the spectral side, and their coefficients [Ar91, §12].
$T_{\mathrm{disc}}(M)$, $T_{\mathrm{ell}}(M)$	Arthur's sets of virtual characters (§4.3).

Symbol	Meaning
$d_M^G(L_1,L_2)$	The constants of the splitting formula (4.5) [Ar88].
W_χ, W_χ°, R_χ, R_χ^M	The stabilizer of χ in W, its reflection subgroup, the R-group, and the R-group relative to M (§3.4).
χ_0	The unramified quadratic character of $SL(2)$, the point $z=-1$ of Example 3.1.
dI_ϕ, dI_γ, dI_ϕ^{sing}, I_B, c_G	The spectral measure of ϕ or of γ, its singular part, the density of its absolutely continuous part, and the constant of Theorem B.
$c_{M,\tau}$, c_τ	Constants from [Ar93] in Theorem 7.1. They are not determined in this paper.
$L_M^\vee$	A lattice in $i\mathfrak{a}_M^*$ such that the twist τ_Λ depends only on Λ modulo $L_M^\vee$ (Theorem 7.1).
$G(T)$, I^T, $I(\chi,\gamma)$, (R), (R′)	Route A: the truncation set, the truncated kernel, its T-independent part, and the regularity hypotheses [JC26, §§3–4, Rem. 2026.20].
δ_B, Q_a, Γ_χ	The modulus character of B, the constant in the volume formula for KaK [JC26, §3.2], and the spherical function.
Q, (U_r), B_r, K_1	Rank-one notation of [JC26, §6]: $Q=1+q^{-1}$, the type of an unramified elliptic element of depth r, the ball of vertices that it fixes, and the first principal congruence subgroup.
κ, H_m, Δ_κ, c_S, $\mathfrak{D}(S)$	For $SL(n)$: an unramified character of $PGL(n,F)$, the elliptic endoscopic group, the transfer factor, a constant, and the finite group that indexes the conjugacy classes in a stable class [JC26, §5].
μ_n, $\mu(\chi)$, T_m, χ_n, M_m	For $SL(n)$: the n-th roots of unity, the roots of unity that preserve the parameter of χ, the set of χ whose parameter is a union of cosets of μ_m, the point T_n, and the Levi subgroup with n/m blocks of size m (§§9.1–9.2).

3.2 Normalized intertwining operators

This subsection recalls the standard intertwining operators between unramified principal series and fixes their normalization. The point is (3.3): the normalized operators fix the spherical vector. Everything in this paper that is special to spherical functions comes from this fact.

For $P,P'\in\mathscr{P}(M_0)$ let $J_{P'|P}(\chi):I_P(\chi)\to I_{P'}(\chi)$ be the standard intertwining operator, defined by a convergent integral for χ in a cone and by meromorphic continuation in general. Gindikin–Karpelevich and Casselman [Ca80, Thm. 3.1] give, in the compact picture,

$$J_{P'|P}(\chi)\Phi_K=\prod_\alpha c_\alpha(\chi)\,\Phi_K,\qquad c_\alpha(\chi)=\frac{1-q^{-1}\chi_\alpha(\pi)}{1-\chi_\alpha(\pi)}=\frac{L(0,\chi_\alpha)}{L(1,\chi_\alpha)},\tag{3.1}$$

the product over the reduced roots α that are positive for P and negative for P'. Define

$$r_{P'|P}(\chi)=\prod_\alpha\frac{L(0,\chi_\alpha)}{L(1,\chi_\alpha)},\qquad R_{P'|P}(\chi)=r_{P'|P}(\chi)^{-1}J_{P'|P}(\chi).\tag{3.2}$$

For unramified data these are Langlands' normalizing factors [La76], with the ε-factors equal to 1 for an additive character of conductor $\mathscr{O}$. They satisfy Arthur's conditions on normalizing factors:

multiplicativity, $R_{P|P'}R_{P'|P}=1$, unitarity and holomorphy on the unitary axis, and compatibility with the K-action. By (3.1),

$$R_{P'|P}(\chi)\Phi_K=\Phi_K \qquad \text{for all } P,P'\in\mathscr{P}(M_0) \text{ and all } \chi. \tag{3.3}$$

Status 3.2. For split G this is standard. Arthur's conditions on normalizing factors, and his canonical normalization [Ar98], have not yet been compared in detail with (3.2); see §12, item (N1).

Example 3.3 (rank one)**.** In the setting of Example 3.1, put $y=\chi_\alpha(\pi)$, so that $y=z$ for $SL(2)$ and $y=z^2$ for $PGL(2)$. There is one pair of distinct parabolic subgroups, and (3.1) reads

$$J_{\bar B|B}(\chi)\Phi_K=c_\alpha(\chi)\Phi_K, \qquad c_\alpha(\chi)=\frac{1-q^{-1}y}{1-y}.$$

On the unit circle c_α has no zero, and it has a pole exactly at $y=1$. This is the point $z=1$ for $SL(2)$, and the two points $z=\pm1$ for $PGL(2)$. By (3.2) the normalizing factor is $r_{\bar B|B}(\chi)=c_\alpha(\chi)$. So the normalized operator $R_{\bar B|B}(\chi)=c_\alpha(\chi)^{-1}J_{\bar B|B}(\chi)$ satisfies $R_{\bar B|B}(\chi)\Phi_K=\Phi_K$ for all χ, which is (3.3). The pole of the unnormalized operator on the spherical vector has been removed.

3.3 Weighted characters

This subsection defines Arthur's weighted characters. They are the terms of the spectral side of his trace formula, and they replace ordinary characters when a Levi subgroup $M\neq G$ is involved. The subsection then proves Lemma 3.4, on which the rest of the paper depends: a weighted character of a spherical function vanishes unless its weight is trivial.

Fix $P\in\mathscr{P}(M_0)$ and unitary χ. For $Q\in\mathscr{P}(M_0)$ and $\Lambda\in i\mathfrak{a}_0^*$ put

$$\mathscr{R}_Q(\Lambda,\chi,P)=R_{Q|P}(\chi)^{-1}R_{Q|P}(\chi_\Lambda),$$

where $\chi_\Lambda=\chi\cdot e^{\Lambda(H_{M_0}(\cdot))}$. Arthur shows that $\{\mathscr{R}_Q(\Lambda,\chi,P)\}_Q$ is a (G,M_0)-family of operators [Ar81, §7]. More generally, for $M\supseteq M_0$ and a Levi $L\supseteq M$ there is an operator $\mathscr{R}_M^L(\chi,P)$, the limit at $\Lambda=0$ of the associated sum over $\mathscr{P}^L(M)$ [Ar81, §6]. The *weighted character* of $f\in C_c^\infty(G(F))$ is

$$J_M(\chi,f)=\operatorname{tr}\big(\mathscr{R}_M(\chi,P)I_P(\chi,f)\big).$$

It does not depend on P. For $M=G$ it is the character $\operatorname{tr}I_P(\chi,f)$. The same construction applies to every tempered representation of a Levi subgroup.

Lemma 3.4. *Let $f\in\mathscr{H}(G,K)$, let χ be unramified, and let $M\subsetneq L$ be Levi subgroups containing M_0. Then $J_M^L(\chi,f)=0$, where J_M^L is formed with the normalization (3.2).*

Proof. Since f is bi-K-invariant, $I_P(\chi,f)$ maps $I_P(\chi)$ into the line $\mathbb{C}\Phi_K$ and kills the K-invariant complement of that line. So $I_P(\chi,f)=f^\vee(\chi)\,e_K$, with e_K the projection onto $\mathbb{C}\Phi_K$, and $J_M^L(\chi,f)=f^\vee(\chi)\langle\mathscr{R}_M^L(\chi,P)\Phi_K,\Phi_K^\vee\rangle$. By (3.3), $\mathscr{R}_Q(\Lambda,\chi,P)\Phi_K=\Phi_K$ for every Q and Λ. So on Φ_K the family is the constant (L,M)-family 1. The limit attached to a constant (L,M)-family is 0 when $M\neq L$, because $\sum_{Q\in\mathscr{P}^L(M)}\theta_Q(\Lambda)^{-1}=0$ [Ar81, Lemma 6.2]. Hence $\mathscr{R}_M^L(\chi,P)\Phi_K=0$. □

Remark 3.5. Lemma 3.4 depends on the normalization. With unnormalized operators, the weighted characters of spherical functions would carry the logarithmic derivatives of the c-functions. These are exactly the terms that appear in the T-independent part of the truncated kernel of Route A; compare §8.

Example 3.6 (a weighted character in rank one)**.** In the setting of Example 3.1, take $M = M_0 = A$ and $P = B$. The (G,A)-family has two members,

$$\mathscr{R}_B(\Lambda,\chi,B) = 1, \qquad \mathscr{R}_{\bar{B}}(\Lambda,\chi,B) = R_{\bar{B}|B}(\chi)^{-1} R_{\bar{B}|B}(\chi_\Lambda).$$

They agree at $\Lambda = 0$, as the members of a (G,A)-family must. Write $\xi = \Lambda(\alpha^\vee)$. With the conventions of [JC26, Example 2.5], $\theta_B(\Lambda) = v^{-1}\xi$ and $\theta_{\bar{B}}(\Lambda) = -v^{-1}\xi$, where v is the volume of $\mathfrak{a}_0/\mathbb{Z}\alpha^\vee$. So

$$\mathscr{R}_A(\chi,B) = \lim_{\xi\to 0} v\,\frac{1 - R_{\bar{B}|B}(\chi)^{-1} R_{\bar{B}|B}(\chi_\Lambda)}{\xi} = -v R_{\bar{B}|B}(\chi)^{-1}\,\frac{d}{d\xi} R_{\bar{B}|B}(\chi_\Lambda)\Big|_{\xi=0}.$$

The operator $\mathscr{R}_A(\chi,B)$ is thus a logarithmic derivative of the normalized intertwining operator, and $J_A(\chi,f) = \operatorname{tr}(\mathscr{R}_A(\chi,B) I_B(\chi,f))$. By contrast, $J_G(\chi,f) = \operatorname{tr} I_B(\chi,f)$ is the ordinary character.

On the spherical vector, Example 3.3 gives $R_{\bar{B}|B}(\chi_\Lambda)\Phi_K = \Phi_K$ for all Λ. The difference quotient is then zero, so $\mathscr{R}_A(\chi,B)\Phi_K = 0$. This is Lemma 3.4 for $M = A$ and $L = G$.

3.4 R-groups and the unramified elliptic set

This subsection recalls the Knapp–Stein R-group of an unramified principal series, and the notion of an elliptic element of it. The R-group controls where $I_P(\chi)$ is reducible. It is needed to describe the singular part of the spectral measure in Theorem C.

For unitary unramified χ let $W_\chi = \{w \in W : w\chi = \chi\}$, and let W_χ° be the subgroup generated by the reflections s_α with $\chi_\alpha = 1$. The Knapp–Stein R-group is $R_\chi = W_\chi / W_\chi^\circ$. It governs the decomposition of $I_P(\chi)$: the commuting algebra is a twisted group algebra of R_χ, which for unramified principal series is untwisted [Ke82]. For Chevalley groups Keys computed R_χ [Ke82]. In particular:

- if the derived group of the dual group $\hat{G}$ is simply connected (equivalently $Z(G)$ is connected), then $R_\chi = 1$ for all unitary unramified χ;
- for $G = SL(n)$, R_χ is cyclic of order dividing n, and $R_\chi \neq 1$ exactly on a finite union of subtori of $\widehat{T}^u$. For $SL(2)$ this is the point χ_0, the unramified quadratic character.

An element $r \in R_\chi$, represented by $w \in W_\chi$, is *elliptic* if w has no non-zero fixed vector in $\mathfrak{a}_0^G$. Arthur's elliptic tempered virtual characters with K-fixed vectors are exactly the virtual characters $\tau = (\chi, r)$ with r elliptic (§6.3).

Example 3.7 (R-groups in rank one)**.** In the setting of Example 3.1, $w\chi = \chi^{-1}$. So $W_\chi = W$ exactly when $z = \pm 1$, and $W_\chi = 1$ otherwise.

Let $G = SL(2)$. For $z = 1$ we have $\chi_\alpha = 1$, so $W_\chi^\circ = W$ and $R_\chi = 1$. For $z = -1$, which is the character χ_0, we have $\chi_\alpha = \mu \neq 1$, so $W_{\chi_0}^\circ = 1$ and $R_{\chi_0} = W$ has order 2. Its non-trivial element

acts by -1 on the line $\mathfrak{a}_0$ and has no non-zero fixed vector, so it is elliptic. This is the only unitary unramified character of $SL(2)$ with a non-trivial R-group, and §10.2 shows what it contributes.

Let $G = PGL(2)$. For $z = \pm 1$ we have $\chi_\alpha(\pi) = z^2 = 1$, so $\chi_\alpha = 1$, $W_\chi^\circ = W$ and $R_\chi = 1$. All R-groups are trivial. This agrees with the first item of the list above, since the center of $PGL(2)$ is trivial.

4 Arthur's local trace formula

This section states the parts of Arthur's local trace formula that we use. We quote; nothing here is new. In Draft 3 the statements of §§4.1–4.2 were compared with [Ar91] and corrected.

4.1 The truncated kernel

This subsection recalls how Arthur defines the distribution $J(f)$. The kernel of a convolution operator is not integrable on the diagonal, so it is truncated first.

The group $G(F) \times G(F)$ acts on $L^2(G(F))$ by $(R(x_1, x_2)\psi)(y) = \psi(x_1^{-1} y x_2)$. For $f = f_1 \otimes f_2$ the operator $R(f)$ has kernel

$$K_f(x,y) = \int_{G(F)} f_1(u) f_2(x^{-1}uy)\, du.$$

Its restriction to the diagonal is not integrable in general. Arthur multiplies it by the characteristic function $u(x,T)$ of a large compact subset of $G(F)$, defined by a point T in the positive chamber through the convex hull of $W \cdot T$. Put $K^T(f) = \int_{G(F)} K_f(x,x) u(x,T)\, dx$. Arthur shows that, as T goes to infinity while staying away from the walls, $K^T(f)$ differs from another function $J^T(f)$ by an exponentially small amount [Ar91, Prop. 4.5]. For p-adic F and T in a lattice, $J^T(f)$ is a finite sum of terms $p_\xi(T,f) e^{\xi(T)}$, in which each p_ξ is a polynomial in T and each $e^{\xi(T)}$ is a root of unity [Ar91, Prop. 6.1]. The constant term $J(f) = p_0(0,f)$ is a distribution on $G(F) \times G(F)$.

4.2 The geometric and spectral expansions

This subsection states the local trace formula. It is an identity between a geometric expression, which is a sum of weighted orbital integrals, and a spectral expression, which is a sum of weighted characters. It also records how Arthur's weighted characters differ from those of §3.3.

Arthur obtains a geometric and a spectral expansion of the constant term $J(f)$ [Ar91, Props. 6.1 and 11.3]. From them he deduces the local trace formula [Ar91, Thm. 12.2]. It is the identity

$$J_{\mathrm{geom}}(f) = J_{\mathrm{spec}}(f)$$

of the two expressions

$$J_{\mathrm{geom}}(f) = \sum_{M \in \mathscr{L}} \frac{|W_0^M|}{|W_0^G|} (-1)^{\dim \mathfrak{a}_M^G} \int_{\Gamma_{\mathrm{ell}}(M)} J_M(\gamma, f)\, d\gamma, \tag{4.1}$$

$$J_{\mathrm{spec}}(f) = \sum_{M \in \mathscr{L}} \frac{|W_0^M|}{|W_0^G|} (-1)^{\dim \mathfrak{a}_M^G} \int_{\Pi_{\mathrm{disc}}(M)} a_{\mathrm{disc}}^M(\pi) J_M(\pi, f)\, d\pi. \tag{4.2}$$

Neither expression is an expansion of $J(f)$ itself. By [Ar91, (12.13)], $J(f)$ is a sum over parabolic subgroups Q of geometric expressions of this kind for the Levi components M_Q, and $J_{\mathrm{geom}}(f)$ is the term $Q = G$. The proof of the formula is an induction on such sums.

Here $\mathscr{L}$ is the set of Levi subgroups containing M_0. $\Gamma_{\mathrm{ell}}(M)$ is the set of conjugacy classes of $M(F)$ whose centralizer in $M(F)$ is compact modulo $A_M(F)$, with the measure of [Ar91, §2]. That measure is carried by the G-regular classes, and it uses the Haar measure of total mass 1 on $T(F)/A_M(F)$ for each elliptic maximal torus T of M. The weighted orbital integral $J_M(\gamma, f)$ is

$$J_M(\gamma,f) = |D(\gamma)| \int_{A_M(F)\backslash G(F)} \int_{A_M(F)\backslash G(F)} f_1(x_1^{-1}\gamma x_1) f_2(x_2^{-1}\gamma x_2)\, v_M(x_1,x_2)\, dx_1\, dx_2,$$

with v_M the weight of the (G,M)-family $e^{-\Lambda(H_P(x_2)-H_{\bar{P}}(x_1))}$ [Ar91, (12.1), (12.2)]. For $M = G$ the weight is 1.

On the spectral side, $\Pi_{\mathrm{disc}}(M)$ is a set of irreducible representations $\pi = \pi_1^\vee \otimes \pi_2$ of $M(F) \times M(F)$. They are the constituents of the representations induced from $\sigma^\vee \otimes \sigma$, where σ is a square-integrable representation of a Levi subgroup M' of M that is fixed by an element of the Weyl group of $(M, A_{M'})$ without non-zero fixed vector in $\mathfrak{a}_{M'}^M$. The coefficients $a_{\mathrm{disc}}^M(\pi)$ and the measure $d\pi$ are defined in [Ar91, (12.4)–(12.6)]. The weighted character is

$$J_M(\pi, f) = \mathrm{tr}\big(\mathscr{J}_M(\pi, P)\, I_P(\pi, f)\big),$$

where $\mathscr{J}_M(\pi,P)$ is the limit of the (G,M)-family $\mathscr{J}_Q(\Lambda,\pi,P) = J_{Q|P}(\pi)^{-1} J_{Q|P}(\pi_\Lambda)$ of *unnormalized* intertwining operators [Ar91, (12.7), (12.8)]. Here $\pi_\Lambda = \pi_{1,-\Lambda}^\vee \otimes \pi_{2,\Lambda}$, and the operator for the product group is

$$J_{Q|P}(\pi_\Lambda) = J_{\bar{Q}|P}(\pi_{1,-\Lambda}^\vee) \otimes J_{Q|P}(\pi_{2,\Lambda}), \tag{4.3}$$

with the opposite parabolic subgroup $\bar{Q}$ in the first factor [Ar91, p. 90].

Remark 4.1 (normalized and unnormalized weighted characters). The weighted characters of §3.3 are formed with the normalized operators $R_{Q|P}$, and Lemma 3.4 needs this. Arthur's $J_M(\pi, f)$ in (4.2) are formed with the unnormalized operators [Ar91, §12, Remark 1]. The two are different, and this remark records what is known about the difference. It is an open point of the paper. The formulas in this remark were derived for Draft 3 from the product formula for (G,M)-families [Ar81, Lemma 6.3], and they have not been compared with the literature.

Write $J_{Q|P} = r_{Q|P} R_{Q|P}$ in both factors of (4.3). Then the unnormalized family is the product of the normalized family and the scalar (G,M)-family

$$r_Q(\Lambda,\pi) = \frac{r_{\bar{Q}|P}(\pi_{1,-\Lambda}^\vee)\, r_{Q|P}(\pi_{2,\Lambda})}{r_{\bar{Q}|P}(\pi_1^\vee)\, r_{Q|P}(\pi_2)}.$$

For $M = M_0$ and $\pi_1 = \pi_2 = \chi$ unramified, (3.2) makes this explicit. Let Σ_P be the set of reduced roots that are positive for P. Then

$$r_Q(\Lambda,\chi) = \prod_{\alpha \in \Sigma_P \cap \Sigma_Q} \frac{c_\alpha(\chi_\Lambda^{-1})}{c_\alpha(\chi^{-1})} \prod_{\alpha \in \Sigma_P \cap \Sigma_{\bar{Q}}} \frac{c_\alpha(\chi_\Lambda)}{c_\alpha(\chi)}. \tag{4.4}$$

In rank one, with $P = B$, this is $r_B(\Lambda,\chi) = c_\alpha(\chi_\Lambda^{-1})/c_\alpha(\chi^{-1})$ and $r_{\bar{B}}(\Lambda,\chi) = c_\alpha(\chi_\Lambda)/c_\alpha(\chi)$. The family is not constant, so the normalizing factors of the two tensor factors do not cancel.

The product formula writes the unnormalized weighted character as a sum over $L \in \mathscr{L}(M)$. The term for L is a scalar $r_M^L(\pi)$, built from logarithmic derivatives of the factors in $r_Q(\Lambda,\pi)$, times a normalized weighted character J_L of the representation induced from π to L. The term $L = M$ has $r_M^M = 1$, and it is the normalized weighted character $J_M(\pi, f)$. The normalized family of π is itself a product over the two tensor factors, indexed by $\bar{Q}$ in the first. On the spherical vector of the first factor it is constant, which is all that Lemma 3.4 uses.

In rank one, $\mathscr{L}(A) = \{A, G\}$. The extra term $L = G$ is $r_A^G(\pi)$ times the ordinary character $\operatorname{tr} I_P(\pi, f)$. For the test function $f \otimes \phi$ of this paper, with ϕ supported on the elliptic set, this character vanishes by Lemma 6.2. So in rank one the two kinds of weighted characters agree on our test functions, and nothing in the rank-one statements of this paper changes.

In rank ≥ 2 the terms with $M \subsetneq L \subsetneq G$ need not vanish, and they have not been computed. By (4.4) the scalars $r_{M_0}^L(\chi)$ are explicit, so this computation is elementary. Until they are, the density in Theorem B and the first term of Theorem 7.1 should be read as follows in rank ≥ 2: they are c_G times Arthur's unnormalized weighted character of $f \otimes \phi$, which is $f^\vee(\chi) J_{M_0}(\chi^{-1}, \phi)$ plus possible terms from the Levi subgroups strictly between M_0 and G. The same applies to the terms with $M \supsetneq M_0$. The smoothness of the total is not affected, since the matrix coefficients of $\mathscr{J}_M(\pi_\lambda, P)$ are smooth in λ [Ar91, Lemma 12.1]. See item (N1) of §12.

Example 4.2 (rank one)**.** Let $G = SL(2)$ or $PGL(2)$. Then $\mathscr{L} = \{A, G\}$. For $M = G$ the coefficient in (4.1) and (4.2) is 1. For $M = A$ we have $|W_0^A| = 1$, $|W_0^G| = 2$ and $\dim \mathfrak{a}_A^G = 1$, so the coefficient is $-\frac{1}{2}$. The geometric side (4.1) therefore reads

$$J_{\mathrm{geom}}(f) = \int_{\Gamma_{\mathrm{ell}}(G)} J_G(\gamma, f)\, d\gamma - \frac{1}{2} \int_{\Gamma_{\mathrm{ell}}(A)} J_A(\gamma, f)\, d\gamma.$$

The first term contains ordinary orbital integrals over the elliptic classes of $G(F)$. The second contains weighted orbital integrals over elements of the split torus $A(F)$, since every element of a torus is elliptic in that torus. The spectral side (4.2) has the same two coefficients. The coefficient $-\frac{1}{2}$ is the constant c_G of Theorem B in rank one.

4.3 The elliptic form

This subsection recalls Arthur's reorganization of the spectral side by virtual characters. It is the form in which the singular part of the spectral measure will appear.

In [Ar93] Arthur reorganizes the spectral side using the virtual characters $\tau \in T_{\mathrm{disc}}(M)$: triples (L, σ, r) with σ a square-integrable representation of a Levi $L \subseteq M$ and r in the R-group of σ in M, up to equivalence. The elliptic ones, $T_{\mathrm{ell}}(M)$, are those for which r is elliptic in the sense of §3.4. The spectral side becomes a sum over M of integrals over $T_{\mathrm{disc}}(M)$, with explicit constants. We need its form for $f = f_1 \otimes f_2$ with f_2 cuspidal (§6).

4.4 Splitting

Our test functions are products $f_1 \otimes f_2$. The splitting formula of this subsection expresses a weighted character of a product through weighted characters of the two factors. In §6 it lets us remove the spherical factor.

For $f = f_1 \otimes f_2$ the weighted characters split [Ar88, Cor. 7.4], [Ar91, §10]:

$$J_M(\pi, f_1 \otimes f_2) = \sum_{L_1, L_2 \in \mathscr{L}(M)} d_M^G(L_1, L_2)\, J_M^{L_1}(\pi, f_{1,Q_1})\, J_M^{L_2}(\pi^\vee, f_{2,Q_2}). \tag{4.5}$$

In (4.5), and from here on, π denotes a representation of $M(F)$. The left side is the weighted character of the representation $\pi \otimes \pi^\vee$ of $M(F) \times M(F)$, which is $\pi_1^\vee \otimes \pi_2$ with $\pi_1 = \pi_2 = \pi^\vee$ in the notation of §4.2. Here $d_M^G(L_1, L_2)$ is non-zero only if $\mathfrak{a}_M^{L_1} \oplus \mathfrak{a}_M^{L_2} = \mathfrak{a}_M^G$, $d_M^G(M, G) = 1$, f_{i,Q_i} are constant terms, and the $Q_i \in \mathscr{P}(L_i)$ are fixed by a choice of a generic point. The same holds for weighted orbital integrals.

The formula is a consequence of the splitting formula for a product of two (G, M)-families [Ar88, Cor. 7.4]. The family attached to $\pi_1 \otimes \pi_2$ is such a product, because its operators act on the two tensor factors separately. Arthur uses the splitting formula in this way in [Ar91, §10]. The argument applies to the normalized operators as well as to the unnormalized ones, and we use (4.5) for the normalized ones. The twist by $-\Lambda$ and the parabolic subgroup $\bar{Q}$ in the first factor of (4.3) could change signs in (4.5). They do not affect the terms that we use. In Proposition 6.1 only the terms with $L_1 = M$ survive. For them no limit is taken in the first factor, whose family has the value 1 at $\Lambda = 0$.

Status 4.3. The statements (4.1) and (4.2) were compared with [Ar91, Thm. 12.2] for Draft 3. The elliptic form of §4.3, the form of (4.5) for weighted characters, and the passage from unnormalized to normalized operators (Remark 4.1) have not been checked against [Ar88, Ar93]. See §12, items (N1) and (N2).

5 The geometric side

This section specializes the geometric side (4.1) to the product of a spherical function and a function supported on the elliptic set. Proposition 5.1 shows that only the term $M = G$ survives, and that it is the distribution $I(f, \phi)$ studied in the companion paper.

Let $f_1 = f \in \mathscr{H}(G, K)$, and let $f_2 = \phi \in C_c^\infty(G(F))$ be supported in the regular elliptic set G_{ell}.

Proposition 5.1. *With* $\mathbf{f} = f \otimes \phi$*:*

(a) $x \mapsto K_{\mathbf{f}}(x, x)$ *has compact support, so the truncated integral* $K^T(\mathbf{f})$ *is constant for* T *large, and*

$$K^T(\mathbf{f}) = \int_{G(F)} \int_{G(F)} f(u)\phi(x^{-1}ux)\, du\, dx = I(f, \phi),$$

the distribution of [JC26, (1.1), (4.8)];

(b) in (4.1) the terms with $M \neq G$ *vanish, and the term* $M = G$ *is the Weyl integration formula for* $I(f, \phi)$*. So* $J_{\mathrm{geom}}(\mathbf{f}) = I(f, \phi)$.

Proof. (a) For ϕ supported on G_{ell}, the set of x with $\phi(x^{-1}ux) \neq 0$ for some u in the support of f is compact, because centralizers of elliptic elements are compact (Kazhdan [Ka86, §4]; [JC26, Lemma 4.4]). So $K_{\mathbf{f}}(x, x)$ has compact support, and $u(x, T) = 1$ on it for T large.

(b) Let $M \neq G$ and $\gamma \in \Gamma_{\mathrm{ell}}(M)$. If γ is regular in G, its centralizer in G is a maximal torus of M, which contains the non-trivial split torus A_M; so γ is not elliptic in G. Hence $\phi(x^{-1}\gamma x) = 0$

for all x, and $J_M(\gamma,\mathbf{f})=0$. The γ that are singular in G form a set of measure zero in $\Gamma_{\mathrm{ell}}(M)$. For $M=G$ the weight is 1, and $J_G(\gamma,\mathbf{f})=|D(\gamma)|O_\gamma(f)O_\gamma(\phi)$ with the measures of [JC26, (4.10)]. Here O_γ denotes the integral over $G(F)$, as in §1. Put $\Phi_\phi(u)=\int_{G(F)}\phi(x^{-1}ux)\,dx$, so that $I(f,\phi)=\int_{G(F)}f(u)\Phi_\phi(u)\,du$. Arthur's form of the Weyl integration formula [Ar91, (2.2)], applied to $h=f\Phi_\phi$, gives $I(f,\phi)=\int_{\Gamma_{\mathrm{ell}}(G)}|D(\gamma)|O_\gamma(f)O_\gamma(\phi)\,d\gamma$, because Φ_ϕ vanishes on the classes that come from $M\neq G$. This is the term $M=G$ of (4.1). □

Example 5.2 (rank one)**.** Let $G=SL(2)$ or $PGL(2)$, and consider the term $M=A$ of Example 4.2. A regular element γ of $A(F)$ has centralizer A, which is split. So γ is not elliptic in G, and neither is any conjugate of it. Since ϕ is supported on G_{ell}, the factor $\phi(x_2^{-1}\gamma x_2)$ in $J_A(\gamma,f\otimes\phi)$ is zero for all x_2. The term with coefficient $-\frac{1}{2}$ therefore vanishes. What remains is the term $M=G$, an integral of $|D(\gamma)|O_\gamma(f)O_\gamma(\phi)$ over the elliptic classes of $G(F)$, and it equals $I(f,\phi)$. So $J_{\mathrm{geom}}(f\otimes\phi)=I(f,\phi)$. This is Proposition 5.1 in rank one.

6 The spectral side

This section specializes the spectral expansion to the same test function $f\otimes\phi$. There are three steps. Splitting removes the weight from the spherical factor (§6.1). Cuspidality of ϕ removes the properly induced data (§6.2). A classification then lists the virtual characters that survive (§6.3).

6.1 Splitting with a spherical and an elliptic function

This subsection applies the splitting formula (4.5) and Lemma 3.4. The result is that the spherical function contributes only its Satake transform.

Proposition 6.1. *Let $f\in\mathscr{H}(G,K)$ and $\phi\in C_c^\infty(G_{\mathrm{ell}})$, and let π be a tempered representation of $M(F)$ induced from a unitary unramified character χ of $M_0(F)$, or more generally any $\tau\in T_{\mathrm{disc}}(M)$. Then*

$$J_M(\pi,f\otimes\phi)=f^\vee(\chi)\,J_M(\pi^\vee,\phi).$$

If π has no $(K\cap M)$-fixed vector, both sides vanish.

Proof. In (4.5), the factor $J_M^{L_1}(\pi,f_{1,\mathcal{Q}_1})$ vanishes for $L_1\neq M$ by Lemma 3.4, applied to the constant term $f_{1,\mathcal{Q}_1}\in\mathscr{H}(L_1,K\cap L_1)$. For $L_1=M$ the condition $\mathfrak{a}_M^M\oplus\mathfrak{a}_M^{L_2}=\mathfrak{a}_M^G$ forces $L_2=G$, and $d_M^G(M,G)=1$. The remaining factor is $J_M^M(\pi,f_{1,\mathcal{Q}})=\operatorname{tr}\pi(f_{1,\mathcal{Q}})=\operatorname{tr}I_{\mathcal{Q}}(\pi,f)$, which equals $f^\vee(\chi)$ if π is a constituent of an unramified principal series with a $K\cap M$-fixed vector, and 0 if it has none. □

6.2 Cuspidality of ϕ

This subsection shows that a function supported on the elliptic set is cuspidal: its trace vanishes on properly induced representations.

Lemma 6.2. *If ϕ is supported on G_{ell}, then $\operatorname{tr}\sigma(\phi)=0$ for every representation σ of finite length induced from a proper parabolic subgroup.*

Proof. By van Dijk's formula [vD72], the character of a properly induced representation is supported on the conjugates of the Levi subgroup, which meet G_{ell} in a set of measure zero. □

So ϕ is cuspidal, and in the elliptic form of the spectral side only elliptic data survive in the ϕ-factor.

6.3 Which virtual characters survive

Combining Proposition 6.1 and Lemma 6.2 with Arthur's elliptic form, and passing to normalized operators as in Remark 4.1, the spectral side $J_{\mathrm{spec}}(f \otimes \phi)$ is a sum over $M \in \mathscr{L}$ of integrals, over the virtual characters $\tau \in T_{\mathrm{disc}}(M)$ with $K \cap M$-fixed vectors, of $f^\vee(\chi_\tau) J_M(\tau^\vee, \phi)$, where χ_τ is the unramified character underlying τ. We now list these τ.

Lemma 6.3. *A square-integrable representation of $L(F)$, L a Levi subgroup of G with $\dim \mathfrak{a}_{M_0}^L > 0$, has no vector fixed by the hyperspecial subgroup $K \cap L(F)$.*

Proof. Write $K_L = K \cap L(F)$, and let A_L be the split part of the center of L. Let σ be an irreducible representation of $L(F)$ that is square-integrable modulo $A_L(F)$, and suppose that it has a non-zero vector v fixed by K_L. Let ω be the character by which $A_L(F)$ acts on σ. It is unitary and unramified.

The Hecke algebra $\mathscr{H}(L, K_L)$ is commutative and σ is irreducible, so the space of K_L-fixed vectors is the line $\mathbb{C}v$, and the Hecke algebra acts on it through a character λ. The matrix coefficient $\Gamma(x) = \langle \sigma(x)v, v \rangle$ is not zero. It is bi-invariant under K_L, it transforms by ω under $A_L(F)$, and it is square-integrable modulo $A_L(F)$. Convolution with $h \in \mathscr{H}(L, K_L)$ multiplies it by a scalar that depends on h through λ. So Γ is a joint eigenvector of the Hecke algebra in the Hilbert space $\mathscr{H}_\omega$ of functions with these three properties.

Macdonald's Plancherel theorem [Ma71] describes $\mathscr{H}_\omega$. Let X_ω be the set of unitary unramified characters of $M_0(F)$ that restrict to ω on $A_L(F)$. It is a coset of a compact torus of dimension $\dim \mathfrak{a}_{M_0}^L$. The Satake transform extends to an isometry of $\mathscr{H}_\omega$ onto the space of W^L-invariant functions in $L^2(X_\omega, \rho_L \, d\chi)$, where $\rho_L \, d\chi$ is absolutely continuous with respect to Haar measure. It carries convolution by h to multiplication by $h^\vee$.

Let F_Γ be the image of Γ. For every h, the function F_Γ vanishes almost everywhere outside the set where $h^\vee$ takes one prescribed value. The Hecke algebra has a countable basis, and the functions $h^\vee$ separate the W^L-orbits, as in the proof of Lemma 8.2. So F_Γ vanishes almost everywhere outside a single orbit. An orbit is finite, and it has measure zero because $\dim \mathfrak{a}_{M_0}^L > 0$. Hence $F_\Gamma = 0$ and $\Gamma = 0$, a contradiction. □

Status 6.4. The proof uses Macdonald's Plancherel theorem for the reductive group $L(F)$ with a fixed central character. We have not checked the form of this statement in [Ma71]. Another proof goes through Casselman's criterion for square-integrability [Ca95]. It needs the additional fact that a tempered spherical representation has a unitary Satake parameter.

By Lemma 6.3, in a triple $\tau = (L, \sigma, r)$ with a K-fixed vector we must have $L = M_0$ and $\sigma = \chi$ a unitary unramified character. So:

- $M = M_0$: $T_{\mathrm{disc}}(M_0) = \widehat{T}^u$, and the contribution is

$$\frac{(-1)^{\dim \mathfrak{a}_0^G}}{|W|} \int_{\widehat{T}^u} f^\vee(\chi) \, J_{M_0}(\chi^{-1}, \phi) \, d\chi.$$

- $M \supsetneq M_0$: $\tau = (M_0, \chi, r)$ with $r \in R_\chi^M$ elliptic in M, i.e. represented by $w \in W_\chi^M$ without non-zero fixed vector in $\mathfrak{a}_0^M$. Unramified twists by $i\mathfrak{a}_M^*$ act on such τ, so each orbit is a torus of dimension $\dim \mathfrak{a}_M^G$ in $\widehat{T}^u/W$. The contribution is an integral of $f^\vee(\chi_\tau) J_M(\tau^\vee, \phi)$ over these tori, with the constant of [Ar93].
- $M = G$: $\tau = (M_0, \chi, r)$ with $r \in R_\chi$ elliptic in G. These τ form a finite set, and contribute point masses $f^\vee(\chi_\tau) \operatorname{tr} \tau^\vee(\phi)$, with constants.

Example 6.5 (rank one)**.** Let $G = PGL(2)$. All R-groups are trivial (Example 3.7), so only the first item of the list occurs. The spectral side $J_{\text{spec}}(f \otimes \phi)$ is

$$-\frac{1}{2} \int_{\widehat{T}^u} f^\vee(\chi) J_A(\chi^{-1}, \phi)\, d\chi.$$

Let $G = SL(2)$. The same term occurs, and the third item contributes as well. By Example 3.7 the only $\tau = (A, \chi, r)$ with r elliptic is the one at $\chi = \chi_0$, and it gives a point mass at χ_0 (§10.2). In rank one there is no Levi subgroup strictly between A and G, so no tori of positive dimension occur.

7 The main theorem

This section equates the geometric side of §5 with the spectral side of §6. The result is Theorem 7.1. Theorems A, B and C of the Introduction follow from it.

Theorem 7.1. *Let $\phi \in C_c^\infty(G_{\text{ell}})$. With the normalizations of §12 assumed, for all $f \in \mathscr{H}(G, K)$,*

$$I(f, \phi) = c_G \int_{\widehat{T}^u} f^\vee(\chi) J_{M_0}(\chi^{-1}, \phi)\, d\chi + \sum_{M \supsetneq M_0} \sum_{[\tau]} c_{M,\tau} \int_{i\mathfrak{a}_M^*/L_M^\vee} f^\vee(\chi_{\tau,\Lambda}) J_M(\tau_\Lambda^\vee, \phi)\, d\Lambda,$$

where $[\tau]$ runs over the finitely many twist-orbits of $\tau = (M_0, \chi, r)$ with r elliptic in M, and the $c_{M,\tau}$ are constants from [Ar93]. Theorems A, B and C follow.

The sum over $M \supsetneq M_0$ includes $M = G$. There $i\mathfrak{a}_G^* = 0$, so the integral is the value at one point, and $J_G(\tau^\vee, \phi) = \operatorname{tr} \tau^\vee(\phi)$. The constant $c_{G,\tau}$ is the one written c_τ in §10.2 and §9.4. These are the terms of the third item in the list of §6.3.

Proof outline. Equate Proposition 5.1 with the spectral side of §6.3.

- Theorem B: the normalized intertwining operators are holomorphic and unitary on the unitary axis, so $\chi \mapsto \mathscr{R}_{M_0}(\chi, P)$ is smooth on $\widehat{T}^u$. Since $I_P(\chi, \phi)$ has finite rank and depends smoothly on χ in the compact picture, $J_{M_0}(\chi^{-1}, \phi)$ is smooth. Divide by ρ to get I_B.
- Theorem C: the same argument on each torus gives smooth densities. The list of §6.3 gives the supports.
- Theorem A: there are finitely many components, each with a smooth density on a compact torus, so the total variation is finite.

□

Status 7.2. The first term of Theorem 7.1 is written with the normalized weighted character J_{M_0}. By Remark 4.1 this is correct in rank one. In rank ≥ 2 further terms from the Levi subgroups strictly between M_0 and G may have to be added, and the same holds for Theorem B. The existence statement (Theorem A), the smoothness of the density, and the description of the supports in Theorem C do not depend on this.

Remark 7.3. Theorem B shows that $I_B\rho$ is always smooth. In Route A, the T-independent part I has double poles on the root walls, canceled by the double zeros of ρ [JC26, Rem. 6.2, (A.4)]. Route B explains this cancellation: it is the smoothness of a weighted character.

Example 7.4 (rank one)**.** Let $G = PGL(2)$. With the same normalizations assumed, Theorem 7.1 and Example 6.5 give

$$I(f,\phi) = -\frac{1}{2}\int_{\widehat{T}^u} f^\vee(\chi)\, J_A(\chi^{-1},\phi)\, d\chi.$$

There is no singular part, and $I_B\rho = -\frac{1}{2}J_A(\chi^{-1},\phi)$. For $G = SL(2)$ there is one more term, a point mass $c_\tau f^\vee(\chi_0)\operatorname{tr}\tau^\vee(\phi)$ at the unramified quadratic character. Both cases are discussed in §§10.1–10.2.

8 Comparison with Route A

This section compares the density I_B of Theorem B with the T-independent part of the truncated kernel of Route A. The comparison is Conjecture 8.1. Proposition 8.3 shows that the two densities agree whenever each route is known to produce a measure, and Corollary 8.5 applies this to $PGL(2)$. We then reformulate the conjecture as a comparison of two operators, explain how the two truncations are related, and say what Hypothesis (R) of the companion paper means from the present point of view.

Conjecture 8.1. *For regular unitary χ, the T-independent part $I(\chi,\phi)$ of the truncated kernel of [JC26, Prop. 4.2] equals $I_B(\chi,\phi)$. Equivalently,*

$$I(\chi,\phi)\,\rho(\chi) = c_G\, J_{M_0}(\chi^{-1},\phi).$$

The displayed form uses Theorem B. In rank ≥ 2 it is subject to Remark 4.1, and the conjecture is then the first statement, $I = I_B$.

Why it should hold. Both I and I_B describe the same linear functional, $f^\vee \mapsto I(f,\phi)$. A measure on $\widehat{T}^u$ that is invariant under W is determined by its integrals against Satake transforms. So whenever each of the two routes produces a measure, the two measures are equal, and so are their absolutely continuous parts. Lemma 8.2 and Proposition 8.3 make this precise. The two hypotheses of the proposition are the conclusions of the two routes.

Lemma 8.2. *The Satake transforms $f^\vee$, $f \in \mathscr{H}(G,K)$, are dense in the space of continuous W-invariant functions on $\widehat{T}^u$, for the uniform norm. Consequently a finite complex W-invariant measure ν on $\widehat{T}^u$ is determined by the numbers $\int f^\vee\, d\nu$, $f \in \mathscr{H}(G,K)$.*

Proof. By the Satake isomorphism [Sa63], the functions $f^\vee$ are exactly the W-invariant finite linear combinations of the characters $\chi \mapsto \chi(\lambda)$ of the compact group $\widehat{T}^u$, where λ runs over $X_*(A)$. They form an algebra that contains the constants. It is closed under complex conjugation, because the conjugate of $\chi(\lambda)$ is $\chi(-\lambda)$.

This algebra separates the W-orbits. Indeed, let $O_1 \neq O_2$ be two orbits. The characters $\chi \mapsto \chi(\lambda)$ separate the points of $\widehat{T}^u$, so by the Stone–Weierstrass theorem there is a finite linear combination p of them that is within $\frac{1}{4}$ of 1 on the finite set O_1 and within $\frac{1}{4}$ of 0 on O_2. The average of p over W is W-invariant, and it takes different values on O_1 and on O_2.

By the Stone–Weierstrass theorem on the compact space $\widehat{T}^u/W$, the algebra is dense. The last statement follows from the Riesz representation theorem, since a W-invariant measure on $\widehat{T}^u$ is the same thing as a measure on $\widehat{T}^u/W$. □

Proposition 8.3. *Let $\phi \in C_c^\infty(G_{\mathrm{ell}})$. Suppose that the following two statements hold.*

(A) There are a W-invariant function $I(\cdot,\phi)$ on $\widehat{T}^u$ with $I(\cdot,\phi)\rho \in L^1(\widehat{T}^u)$, and a finite complex W-invariant measure ν_A on $\widehat{T}^u$ that is singular with respect to Haar measure, such that

$$I(f,\phi) = \int_{\widehat{T}^u} f^\vee(\chi) I(\chi,\phi)\rho(\chi)\, d\chi + \int_{\widehat{T}^u} f^\vee \, d\nu_A \qquad \text{for all } f \in \mathscr{H}(G,K).$$

(B) The same holds with a function $I_B(\cdot,\phi)$ and a measure ν_B.

Then $I(\chi,\phi)\rho(\chi) = I_B(\chi,\phi)\rho(\chi)$ for almost every $\chi \in \widehat{T}^u$, and $\nu_A = \nu_B$.

Proof. The measures $I(\cdot,\phi)\rho\, d\chi + \nu_A$ and $I_B(\cdot,\phi)\rho\, d\chi + \nu_B$ are finite, complex and W-invariant, and they have the same integral against every $f^\vee$. By Lemma 8.2 they are equal. The decomposition of a finite complex measure into a part that is absolutely continuous with respect to Haar measure and a part that is singular is unique. So the two absolutely continuous parts are equal, which gives the first statement, and the two singular parts are equal. □

When the hypotheses hold. Hypothesis (B) is the conclusion of Theorem 7.1. There ν_B is the sum of the terms with $M \supsetneq M_0$. Each of them is carried by a torus of dimension smaller than that of $\widehat{T}^u$, so ν_B is singular. The function $I_B\rho$ is the density of the first term. In rank one it is W-invariant by Lemma 8.4. In general it can be made W-invariant by averaging over W, which does not change the first term of Theorem 7.1, because $f^\vee$ is W-invariant.

Hypothesis (A) is the conclusion of Route A in the following cases.

- Under the hypotheses (C) and (R) of [JC26] for G and for the level n of ϕ, with $\nu_A = 0$ [JC26, Thm. 4.5(b)]. The weaker (R′) of [JC26, Rem. 2026.20] suffices. Here $I(\cdot,\phi)$ is the T-independent part of the truncated kernel. It is W-invariant, because the truncated kernel $I^T(\chi,\phi)$ depends on χ only through the spherical function, and the expansion of [JC26, Prop. 4.2(c)] is unique.
- For $G = PGL(2)$ and every level, with $\nu_A = 0$. In rank one (C) is automatic, and (R) holds by [JC26, Lemma 6.1A].
- For $G = SL(2)$ and the orbital integral of an unramified elliptic γ of depth 0, with $\nu_A = \frac{1}{2}\delta_{\chi_0}$ [JC26, Prop. 6.4].

In general, Hypothesis (A) is [JC26, Conj. 4.6(a)]. For $PGL(3)$ it is supported by the numerical results of [JC26, App. A], with $\nu_A = 0$.

Lemma 8.4. *Let χ be a unitary unramified character, $w \in W$ and $f \in C_c^\infty(G(F))$. Then $J_{M_0}(w\chi, f) = J_{M_0}(\chi, f)$.*

Proof. Since G is split and $K = G(\mathscr{O})$, the element w has a representative $\tilde{w}$ in K. For $P \in \mathscr{P}(M_0)$ put $wP = \tilde{w}P\tilde{w}^{-1}$. Left translation, $(T\varphi)(g) = \varphi(\tilde{w}^{-1}g)$, is an isomorphism of $I_P(\chi)$ onto $I_{wP}(w\chi)$ that commutes with the action of $G(F)$. So $T\, I_P(\chi, f)\, T^{-1} = I_{wP}(w\chi, f)$. It preserves the compact picture, because $\tilde{w} \in K$.

The integral that defines $J_{Q|P}(\chi)$ is carried by T to the integral that defines $J_{wQ|wP}(w\chi)$, and conjugation by $\tilde{w}$ preserves the Haar measures normalized by K. So $T\, J_{Q|P}(\chi)\, T^{-1} = J_{wQ|wP}(w\chi)$. The roots that are positive for wP and negative for wQ are the $w\alpha$, with α positive for P and negative for Q, and $(w\chi)_{w\alpha} = \chi_\alpha$. By (3.2), $r_{wQ|wP}(w\chi) = r_{Q|P}(\chi)$. Hence $T\, R_{Q|P}(\chi)\, T^{-1} = R_{wQ|wP}(w\chi)$.

Since $(w\chi)_{w\Lambda} = w(\chi_\Lambda)$, it follows that $T\, \mathscr{R}_Q(\Lambda, \chi, P)\, T^{-1} = \mathscr{R}_{wQ}(w\Lambda, w\chi, wP)$. Also $\theta_{wQ}(w\Lambda) = \theta_Q(\Lambda)$. Summing over Q and taking the limit gives $T\, \mathscr{R}_{M_0}(\chi, P)\, T^{-1} = \mathscr{R}_{M_0}(w\chi, wP)$. Taking traces,

$$J_{M_0}(\chi, f) = \operatorname{tr}\big(\mathscr{R}_{M_0}(w\chi, wP)\, I_{wP}(w\chi, f)\big) = J_{M_0}(w\chi, f).$$

The last step uses that the weighted character does not depend on the parabolic subgroup (§3.3). □

Corollary 8.5. *Let $G = PGL(2)$, and assume Theorem 7.1 for G. Then Conjecture 8.1 holds for G: for every $\phi \in C_c^\infty(G_{\mathrm{ell}})$ that is bi-invariant under a principal congruence subgroup,*

$$I(\chi, \phi)\, \rho(\chi) = -\tfrac{1}{2} J_A(\chi^{-1}, \phi) \qquad \text{for almost every } \chi \in \widehat{T}^u.$$

Proof. All R-groups of $PGL(2)$ are trivial (Example 3.7), so Theorem 7.1 has no term with $M \supsetneq M_0$ and $\nu_B = 0$. In rank one the density of the first term is $c_G J_A(\chi^{-1}, \phi) = -\frac{1}{2} J_A(\chi^{-1}, \phi)$ (Remark 4.1 and Example 7.4). It is W-invariant by Lemma 8.4. Hypothesis (A) holds with $\nu_A = 0$, by the second case above. Now apply Proposition 8.3. □

For $SL(2)$ and γ as in the third case, the same argument gives more. If Theorem 7.1 holds, then the two densities agree, and the two point masses at χ_0 agree as well. This is how §10.2 fixes the constant c_τ.

What the argument does not show. Proposition 8.3 does not prove Conjecture 8.1, for two reasons. First, Hypothesis (A) is known only in the cases listed. Second, the argument is indirect even where both hypotheses hold. It identifies the two densities because they represent the same functional. It does not explain why the constant term of a truncated kernel should be a weighted character. A direct proof would compare the two constructions (item (N6) of §12).

A reformulation. The two sides of Conjecture 8.1 have the same shape, and this suggests how a direct proof could go. Let ϕ be bi-invariant under the principal congruence subgroup K_n, and let χ be regular. By [JC26, Prop. 4.2(a)],

$$I^T(\chi, \phi) = \sum_{i,j} J^T(\chi, \chi^{-1}, u_i^*, u_j)\, \langle \pi_\chi(\phi) u_i, u_j^* \rangle,$$

where $\{u_i\}$ is a basis of the K_n-fixed vectors of the principal series π_χ, $\{u_i^*\}$ is the dual basis, and the J^T are truncated inner products of matrix coefficients. The coefficients $J^T(\chi,\chi^{-1},u_i^*,u_j)$ do not involve ϕ. Under (C) each of them is a finite sum of polynomials in T times characters of T [JC26, Prop. 3.6]. Let $\mathscr{J}_{ij}(\chi)$ be the constant coefficient of the trivial character. Then, by the definition in [JC26, Prop. 4.2(c)],

$$I(\chi,\phi)=\sum_{i,j}\mathscr{J}_{ij}(\chi)\,\langle\pi_\chi(\phi)u_i,u_j^*\rangle=\operatorname{tr}\big(\mathscr{J}(\chi)\,\pi_\chi(\phi)\big)$$

for an operator $\mathscr{J}(\chi)$ on the K_n-fixed vectors. On the other side, at least in rank one,

$$I_B(\chi,\phi)\,\rho(\chi)=c_G\operatorname{tr}\big(\mathscr{R}_{M_0}(\chi^{-1},P)\,I_P(\chi^{-1},\phi)\big).$$

So the conjecture compares two operators on the vectors of level n in an unramified principal series, through their traces against the operators $\pi_\chi(\phi)$ with ϕ supported on the elliptic set. Both operators are constant terms. The first is the constant term of a truncated inner product, and the second is the limit of a (G,M)-family. The rank-one test of item (N5) is a comparison of these two operators on the $(q+1)$-dimensional space of vectors of level one.

The two truncations. Arthur's $u(x,T)$ and the set $G(T)$ of Route A are both defined by the convex hull of $W\cdot T$. Route A's vertex terms $\psi(T_Q)R_Q$ [JC26, Lemma 3.5A] are periodic versions of Arthur's $e^{\Lambda(T_Q)}c_Q(\Lambda)/\theta_Q(\Lambda)$. In rank one, the constant term at $\Lambda=0$ of $\sum_Q e^{\Lambda(T_Q)}c_Q(\Lambda)\theta_Q(\Lambda)^{-1}$ is a logarithmic derivative of an intertwining operator. This is the mechanism that produces a weighted character, and it is visible in [JC26, (6.5)].

What (R) means here. Hypothesis (R) of [JC26] asks that each oscillatory term of I^T, times ρ, be smooth. Theorem C says that only elliptic data on non-trivial R-groups can leave a singular part. For $PGL(3)$ all R-groups are trivial, and the poles found in [JC26, App. A] on the subtori $z_i^3=1$ must cancel in the limit, as observed. The weaker condition (R′) of [JC26, Rem. 2026.20] is a candidate for the analytic shadow of "no elliptic data".

Example 8.6 ($PGL(2)$)**.** For $G=PGL(2)$ and γ of type (U_r), Route A gives $I(\chi,\gamma)\rho(\chi)$ in closed form. The formula is displayed in §10.1. Conjecture 8.1 says that this explicit function equals $-\frac{1}{2}J_A(\chi^{-1},\phi)$ for ϕ concentrated near γ. By Example 3.6, the second function is a trace against a logarithmic derivative of an intertwining operator. The numerical test of this equality is item (N5) of §12.

9 $SL(n)$ and endoscopy

This section studies $G=SL(n)$, where the singular part of the spectral measure should be expressed by κ-orbital integrals and endoscopic groups. We first make both descriptions explicit. On the side of Route B we compute the R-groups (§9.1) and the elliptic data (§9.2). On the side of Route A we compute the supports of the κ-orbital integrals (§9.3). The two index sets and the two families of supports turn out to be the same (Proposition 9.6). We then say what Theorem D asserts and what is missing (§9.4). The identification of the densities remains an outline and not a proof.

For $G = SL(n)$ and κ an unramified character of $PGL(n,F)$ of order $m \mid n$, the κ-orbital integrals and the elliptic endoscopic groups H_m are as in [JC26, §5]. Arthur's local character relations [Ar96] and his transfer of weighted orbital integrals [Ar99] relate the elliptic virtual characters $\tau = (M_0, \chi, r)$ of §6.3 to the stable characters of the H_m: the elliptic element $r \in R_\chi$ corresponds to a κ of order equal to that of r. Combined with Theorem 7.1, this would prove Theorem D and remove the hypothesis of [JC26, Prop. 5.2]. The stabilization of the local trace formula (Arthur's stable trace formula papers; see also Peng's explicit stable local trace formula [Pe16]) is the natural framework.

9.1 Unramified characters and R-groups of $SL(n)$

This subsection makes the R-groups of §3.4 explicit for $SL(n)$. The result is Lemma 9.1: the R-group of χ is the group of roots of unity that preserve the parameter of χ.

Let $G = SL(n)$, with A the diagonal torus and $W = S_n$. The dual group is $PGL(n,\mathbb{C})$. A unitary unramified character χ of $A(F)$ is given by a parameter $z = (z_1, \dots, z_n)$ with $|z_i| = 1$, taken up to a common scalar factor:

$$\chi\big(\operatorname{diag}(\pi^{k_1}, \dots, \pi^{k_n})\big) = \prod_i z_i^{k_i} \qquad \Big(\sum_i k_i = 0\Big).$$

For the root $\alpha = e_i - e_j$ we have $\chi_\alpha(\pi) = z_i/z_j$. The Weyl group permutes the coordinates, $(w\cdot z)_i = z_{w^{-1}(i)}$. For $n = 2$ the coordinate of Example 3.1 is z_1/z_2. Write μ_n for the group of n-th roots of unity, and put

$$\mu(\chi) = \{\zeta \in \mathbb{C}^\times : \ (\zeta z_1, \dots, \zeta z_n) \text{ is a permutation of } (z_1, \dots, z_n)\}.$$

This set does not change when z is multiplied by a scalar.

Lemma 9.1. *Let χ be a unitary unramified character of $SL(n)$, with parameter z.*

(a) $\mu(\chi)$ is a subgroup of μ_n.

(b) For $w \in W_\chi$ there is a unique $c(w) \in \mathbb{C}^\times$ with $w \cdot z = c(w)z$. The map $c : W_\chi \to \mu(\chi)$ is a surjective homomorphism with kernel W_χ°. So $R_\chi \cong \mu(\chi)$, a cyclic group of order dividing n.

(c) For $m \mid n$, the group R_χ contains μ_m if and only if the multiset $\{z_1, \dots, z_n\}$ is a union of cosets of μ_m in the unit circle.

Proof. (a) If ζz is a permutation of z, then the product of the coordinates gives $\zeta^n \prod z_i = \prod z_i$, so $\zeta^n = 1$. The set $\mu(\chi)$ is closed under products and inverses.

(b) The character $w\chi$ has parameter $w \cdot z$, and two parameters give the same character exactly when they differ by a scalar. So $w \in W_\chi$ means $w \cdot z = cz$ for a scalar c, which is unique and lies in $\mu(\chi)$. Every $\zeta \in \mu(\chi)$ arises, by the definition of $\mu(\chi)$. If $w \cdot z = cz$ and $w' \cdot z = c'z$ then $(ww') \cdot z = w \cdot (c'z) = cc'z$, so c is a homomorphism. Its kernel is the stabilizer of the vector z in S_n. This is the group of permutations that preserve each set of equal coordinates, and it is generated by the transpositions $(i\,j)$ with $z_i = z_j$. These are the reflections s_α with $\chi_\alpha = 1$, so the kernel is W_χ°.

(c) By (b), $R_\chi \supseteq \mu_m$ means that multiplication by each $\zeta \in \mu_m$ preserves the multiset $\{z_i\}$. This holds exactly when the multiset is a union of cosets of μ_m, counted with multiplicity. □

For $m \mid n$ with $m > 1$, put $d = n/m$ and let $T_m \subset \widehat{T}^u/W$ be the set of χ whose parameter is a union of d cosets $w_1\mu_m, \dots, w_d\mu_m$. It is the image of a torus of dimension $d-1$, with coordinates $(w_1, \dots, w_d)$ up to a scalar. If m' divides m then $T_m \subseteq T_{m'}$. By Lemma 9.1(c), $R_\chi \neq 1$ exactly on the union of the T_m. The set T_n is a single point χ_n, with parameter $(1, \zeta_n, \dots, \zeta_n^{n-1})$ for a primitive n-th root of unity ζ_n, and $R_{\chi_n} = \mu_n$. For $n = 2$ it is χ_0, and for $n = 3$ it is the point χ_ζ of §10.4. This proves, for $SL(n)$, the statements quoted from [Ke82] in §3.4.

9.2 The elliptic data

This subsection determines, for $SL(n)$, the virtual characters that occur in the singular part of Theorem 7.1. They are the triples (M_0, χ, r) with r elliptic in a Levi subgroup $M \supsetneq M_0$.

A convention. An element r of R_χ^M is a coset of $W_\chi^{M,\circ}$ in W_χ^M. Following Arthur, we represent it by the unique element of the coset that preserves the set of positive roots α of M with $\chi_\alpha = 1$ [Ar91, §12, Remark 3]. Ellipticity of r in M refers to this representative. When $W_\chi^{M,\circ} = 1$ there is nothing to choose. The definition in §3.4 does not say which representative is meant, and the choice matters. For $SL(4)$ and $z = (1, 1, -1, -1)$, the class with $c = -1$ contains the 4-cycle $(1\,3\,2\,4)$, which has no non-zero fixed vector in $\mathfrak{a}_0$. Its representative in the sense above is $(1\,3)(2\,4)$, which has one, so this class is not elliptic in G.

Up to conjugation by W, a Levi subgroup $M \supseteq M_0$ of $SL(n)$ consists of the block-diagonal matrices of determinant 1 with blocks of sizes $n_1, \dots, n_k$. Then $W^M = S_{n_1} \times \cdots \times S_{n_k}$, and $\mathfrak{a}_0^M$ is the space of vectors in $\mathbb{R}^n$ whose coordinates add up to zero on each block. For $m \mid n$ and $d = n/m$, let M_m be the Levi subgroup with d blocks of size m. For $m = n$ it is G.

Proposition 9.2. *Let $G = SL(n)$, let $M \supsetneq M_0$ be a block-diagonal Levi subgroup, let χ be a unitary unramified character with parameter z, and let $r \in R_\chi^M$ be elliptic in M. Then there are a divisor $m > 1$ of n and a primitive m-th root of unity ζ such that:*

(a) $M = M_m$.

(b) On each block, the coordinates of z form one coset of μ_m, so $\chi \in T_m$.

(c) The element r is the class of a permutation w that is a single m-cycle on each block, and $c(w) = \zeta$. For given χ and ζ there is only one such w.

Conversely, every triple (M_m, χ, r) as in (a)–(c) has r elliptic in M_m. The twists of such a χ by $i\mathfrak{a}_{M_m}^*$ fill out T_m, and $\dim T_m = n/m - 1 = \dim \mathfrak{a}_{M_m}^G$.

Proof. Let $w \in W^M$ be the representative of r fixed by the convention, and let $\zeta = c(w)$, so that $z_{w^{-1}(i)} = \zeta z_i$ for all i. The space of vectors in $\mathfrak{a}_0^M$ fixed by w has dimension $\sum_i (c_i - 1)$, where c_i is the number of cycles of w in the i-th block. So r is elliptic in M exactly when w is a single n_i-cycle on each block.

Assume this. Going once around the cycle of the i-th block gives $\zeta^{n_i} = 1$. Let m be the order of ζ. Then m divides every n_i.

If $m = 1$ then w fixes z, so $w \in W_\chi^{M,\circ}$ and r is the trivial class. Its representative is the identity, which is a single cycle on each block only if all $n_i = 1$. This is excluded by $M \neq M_0$. So $m > 1$.

Suppose that $n_i > m$ for some i, and let j be an index in the i-th block. Then $w^{-m}(j) \neq j$ and $z_{w^{-m}(j)} = \zeta^m z_j = z_j$. So $\beta = e_j - e_{w^{-m}(j)}$ is a root of M with $\chi_\beta = 1$. Replacing β by $-\beta$ if

necessary, we may assume that β is positive. For every k, the root $w^k\beta = e_{w^k(j)} - e_{w^{k-m}(j)}$ again satisfies $\chi_{w^k\beta} = 1$, because $z_{w^k(j)} = \zeta^{-k} z_j = z_{w^{k-m}(j)}$. It is positive, because w preserves the positive roots α of M with $\chi_\alpha = 1$. But

$$\sum_{k=0}^{n_i-1} w^k\beta = \sum_{k=0}^{n_i-1} e_{w^k(j)} - \sum_{k=0}^{n_i-1} e_{w^{k-m}(j)} = 0,$$

since both sums run over the whole block. A sum of positive roots is not zero. This contradiction shows that $n_i = m$ for all i, which is (a). On each block the coordinates are then $z_j, \zeta z_j, \dots, \zeta^{m-1} z_j$ in the order of the cycle, a coset of μ_m. This gives (b) and the first statement of (c). The coordinates in a block are distinct, so the relation $z_{w^{-1}(i)} = \zeta z_i$ determines $w^{-1}(i)$. Hence w is unique. For the same reason $W_\chi^{M_m,\circ} = 1$ for every χ as in (b), even when two blocks carry the same coset, because the roots of M_m compare coordinates within one block only.

Conversely, let (M_m, χ, r) be as in (a)–(c). The coordinates of z in a block are distinct, so $W_\chi^{M_m,\circ} = 1$ and w is the representative of r. It is a single cycle on each block, so r is elliptic in M_m. A twist by $i\mathfrak{a}_{M_m}^*$ multiplies the coordinates of the j-th block by a common scalar u_j of absolute value 1. So the twists move the cosets $w_j\mu_m$ independently, and they fill out T_m. Finally $\mathfrak{a}_{M_m}^G$ is the space of vectors that are constant on each of the d blocks and have sum zero, of dimension $d-1$. □

Corollary 9.3. *For $G = SL(n)$, the terms with $M \supsetneq M_0$ in Theorem 7.1 are indexed, up to conjugation by W, by the pairs (m,ζ) with $m \mid n$, $m > 1$ and ζ a primitive m-th root of unity. Equivalently, they are indexed by the elements $\zeta \neq 1$ of μ_n. The term for (m,ζ) is carried by T_m. The terms with $M = G$ are those with $m = n$. They are the $\varphi(n)$ virtual characters $\tau_\zeta = (M_0, \chi_n, r_\zeta)$, ζ a primitive n-th root of unity, and they all sit at the one point χ_n.*

Proof. This is Proposition 9.2 applied to the list of §6.3. □

In particular, if n is prime then the singular part consists of point masses at χ_n only.

Example 9.4 ($n = 4$)**.** For $SL(4)$ there are three elements $\zeta \neq 1$ of μ_4. For $\zeta = -1$ we have $m = 2$ and $M_2 = S(GL(2) \times GL(2))$. The set T_2 consists of the parameters $(a, -a, b, -b)$, and it has dimension $1 = \dim \mathfrak{a}_{M_2}^G$. The permutation $w = (1\,2)(3\,4)$ multiplies such a parameter by -1, and it is elliptic in M_2. For $\zeta = \pm i$ we have $m = 4$ and $M_4 = G$, and both terms sit at the point χ_4 with parameter $(1, i, -1, -i)$. Here $R_{\chi_4} = \mu_4$. Its two elements of order 4 are elliptic in G. Its element -1 is not elliptic in G, but χ_4 lies on T_2, and there -1 is elliptic in a conjugate of M_2.

9.3 The side of Route A: κ-orbital integrals

This subsection recalls from [JC26, §5] how an orbital integral on $SL(n)$ decomposes into κ-orbital integrals. It then determines where each piece is supported on the spectral side.

The unramified characters κ of $PGL(n,F)$ that are trivial on the image of $SL(n,F)$ are the characters $\kappa_\zeta = \eta \circ \det$, where η is the unramified character of $F^\times$ with $\eta(\pi) = \zeta$ and $\zeta \in \mu_n$ [JC26, §5.1]. The order of κ_ζ is the order m of ζ, and its endoscopic group is H_m. So the non-trivial κ are indexed by the elements $\zeta \neq 1$ of μ_n.

Let γ be regular elliptic with centralizer S, and let $f \in \mathscr{H}(G,K)$. In this subsection and the next we use the notation and the normalizations of [JC26, §5]. In particular O_γ and O_s denote integrals over $S(F)\backslash G(F)$, and not over $G(F)$ as in §1. The two differ by the factor $\mathrm{vol}(S(F))$. The transfer factors Δ_κ have absolute value 1 [JC26, §5.2]. The formulas (5.4A) and (5.11) and Lemma 5.0 of that paper give

$$O_\gamma(f) = \frac{1}{c_S|\mathfrak{D}(S)|}\,\Phi^1_f(\gamma) + \frac{1}{|\mathfrak{D}(S)|}\sum_{\kappa\neq 1}\Delta_\kappa(\gamma_H,\gamma)^{-1}\,S\Phi^H_{\iota^* f}(\gamma_H). \tag{9.1}$$

The sum is over the non-trivial unramified characters κ of $\mathfrak{D}(S)$ such that γ comes from an element γ_H of $H = H_m$, with m the order of κ. The first term is the stable part. Note that the constant c_S cancels in the terms with $\kappa \neq 1$. Since $(\iota^* f)^\vee = f^\vee \circ \iota$, the term for κ depends on f only through the restriction of $f^\vee$ to the image of ι.

Lemma 9.5. *Let $m \mid n$, $m > 1$ and $d = n/m$. Let $\iota : {}^L H_m \to {}^L G$ be the L-embedding that is block-diagonal on the dual group of H_m and sends the Frobenius element to the cyclic permutation of the m blocks. Then the image of $\widehat{T}^u_{H_m}/W_{H_m}$ under ι is T_m.*

Proof sketch. The group H_m is the kernel of $N_{E/F}\circ\det$ on $\mathrm{Res}_{E/F}\,GL(d)$, with E/F unramified of degree m. Its dual group is $GL(d,\mathbb{C})^m$ modulo the scalars embedded diagonally, and the Frobenius element σ permutes the m factors cyclically. The unramified characters of $H_m(F)$, up to W_{H_m}, correspond to the semisimple classes of elements $(g_1,\dots,g_m)\rtimes\sigma$ under conjugation by the dual group. Such a class is determined by the conjugacy class of the product $g_1 g_2\cdots g_m$ in $GL(d,\mathbb{C})$, up to multiplication by the m-th power of a scalar. Let $u_1,\dots,u_d$ be the eigenvalues of this product. The image under ι is the class in $PGL(n,\mathbb{C})$ of a block-cyclic matrix whose m-th power is block-diagonal, with blocks conjugate to the product. Its eigenvalues are therefore all the m-th roots of $u_1,\dots,u_d$. This multiset is a union of d cosets of μ_m, and every such union arises. The unitary characters correspond to $|u_j| = 1$. □

For $m = n$ we have $d = 1$, the group H_n is the norm-one torus of E, and the lemma gives the point χ_n, which is the χ_κ of [JC26, Prop. 5.1]. The paper [JC26] fixes an L-embedding without specifying it. We assume that it is the one of the lemma.

Proposition 9.6 (the dictionary)**.** *Let $G = SL(n)$.*

(a) The non-trivial unramified characters κ_ζ and the terms with $M \supsetneq M_0$ of Theorem 7.1 have the same index set, the elements $\zeta \neq 1$ of μ_n.

(b) Let ζ have order m. The term for κ_ζ in (9.1) depends only on the restriction of $f^\vee$ to T_m. The term for (m,ζ) in Theorem 7.1 is carried by the same set T_m.

(c) T_m is the set of χ whose R-group contains μ_m.

Proof. (a) and the second statement of (b) are Corollary 9.3. The first statement of (b) is Lemma 9.5. (c) is Lemma 9.1(c). □

This makes precise the statement, at the beginning of this section, that the elliptic element r corresponds to a κ of the same order.

9.4 What Theorem D asserts, and what remains

This subsection states Theorem D in terms of the dictionary. It shows that a summed form of the theorem follows from Theorem 7.1 and from statements of Route A, writes out the case of prime n, and lists the missing steps.

By (9.1), integrated against ϕ as in [JC26, (4.11)], Route A writes the functional $f^\vee \mapsto I(f,\phi)$ as a stable term, which comes from $\kappa = 1$, plus one term for each $\zeta \neq 1$ in μ_n. The term for ζ is a functional of the restriction of $f^\vee$ to T_m. Route B writes the singular part of the same functional as a sum over the same ζ of measures on the sets T_m. Theorem D asserts that the two sums agree. The uniqueness argument of §8 reduces this to statements about each route separately.

Proposition 9.7. *Let $G = SL(n)$ and $\phi \in C_c^\infty(G_{\mathrm{ell}})$. Assume the following.*

(i) Theorem 7.1 holds for G and ϕ.

(ii) The stable term of Route A is given by an absolutely continuous measure on $\widehat{T}^u$.

(iii) For each $\zeta \neq 1$ in μ_n, the term of Route A for κ_ζ is given by a finite complex measure ν_ζ on $\widehat{T}^u/W$ carried by T_m.

Then the singular part of the spectral measure in Theorem 7.1 is $\sum_{\zeta\neq 1}\nu_\zeta$. In particular it is a sum of push-forwards from the endoscopic groups whenever each ν_ζ is, as in [JC26, Prop. 5.2].

Proof. Each T_m is the image of a torus of dimension smaller than that of $\widehat{T}^u$, so $\sum_\zeta \nu_\zeta$ is singular with respect to Haar measure. By (ii) and (iii), Hypothesis (A) of Proposition 8.3 holds with $\nu_A = \sum_\zeta \nu_\zeta$. By (i), Hypothesis (B) holds, with ν_B the singular part of Theorem 7.1. Proposition 8.3 gives $\nu_B = \nu_A$. □

Assumption (iii) holds without further hypothesis for ζ of order n, where ν_ζ is a point mass at χ_n [JC26, Prop. 5.1]. For ζ of order $m < n$ it follows from a stable form of the spectral identity for H_m [JC26, Prop. 5.2], which is not known. Assumption (ii) is not known either.

Proposition 9.7 identifies the sum of the singular terms and not the individual terms. Different ζ of the same order share the set T_m, and $T_m \subseteq T_{m'}$ when m' divides m. So the supports alone do not separate the terms. A term-by-term statement needs the character identities of [Ar96].

Prime n. Let n be prime. Then every $\zeta \neq 1$ has order n, and both descriptions of the singular part are point masses at χ_n. Let S_n be the norm-one torus of the unramified extension E of degree n. For a torus S stably conjugate to S_n the group $\mathfrak{D}(S)$ is isomorphic to $F^\times/N_{E/F}E^\times$, which is cyclic of order n, and all its characters are unramified. For such S and regular $\gamma \in S(F)$, the formula (9.1) and [JC26, Prop. 5.1] give

$$O_\gamma(f) = \frac{1}{nc_S}\Phi_f^1(\gamma) + \Big(\frac{1}{n}\sum_{\kappa\neq 1}\Delta_\kappa(\gamma_H,\gamma)^{-1}\Big) f^\vee(\chi_n).$$

For elliptic γ in other tori no term with $\kappa \neq 1$ occurs, by [JC26, (5.11)]. If all transfer factors are 1, the weight of the point mass is $\frac{n-1}{n}$. This is $\frac{1}{2}$ for $SL(2)$ and $\frac{2}{3}$ for $SL(3)$, as in §§10.2–10.4. On the side of Route B, the terms with $M = G$ of Theorem 7.1 are $\sum_{\zeta\neq 1} c_{\tau_\zeta} f^\vee(\chi_n)\,\mathrm{tr}\,\tau_\zeta^\vee(\phi)$. Comparing

the coefficients of $f^\vee(\chi_n)$, and integrating over γ with [JC26, (4.11)], we find that for prime n the conclusion of Proposition 9.7 is the identity

$$\sum_{\zeta\neq 1} c_{\tau_\zeta}\,\mathrm{tr}\,\tau_\zeta^\vee(\phi) = \sum_S \frac{\mathrm{vol}(S(F))}{|W(G,S)|}\int_{S(F)} |D(s)|\Big(\frac{1}{n}\sum_{\kappa\neq 1}\Delta_\kappa(s_H,s)^{-1}\Big)O_s(\phi)\,ds. \tag{9.2}$$

Here S runs over the conjugacy classes of tori that are stably conjugate to S_n, and the measures and orbital integrals on the right are those of [JC26, §§4–5]. By the Weyl integration formula, (9.2) is a statement about characters. It says that, on the regular elliptic set, the virtual character $\sum_{\zeta\neq 1} c_{\tau_\zeta}\Theta_{\tau_\zeta^\vee}$ vanishes off the tori stably conjugate to S_n, and that on such a torus it equals $\mathrm{vol}(S(F))\,\frac{1}{n}\sum_{\kappa\neq 1}\Delta_\kappa(s_H,s)^{-1}$. For $n=2$ this is the behavior of the character of $\pi^+-\pi^-$ that enters §10.2. The identity (9.2) has not been proved here. Its constants depend on the normalizations of [JC26, §5], which have been checked only for $SL(2)$.

What remains. Three steps are missing for Theorem D.

(i) Theorem 7.1 itself, with its constants $c_{M,\tau}$ (items (N1), (N2) and (N4) of §12).

(ii) Assumptions (ii) and (iii) of Proposition 9.7 for ζ of order less than n. They are stable spectral identities for G and for the groups H_m.

(iii) The term-by-term identification of the densities, from the local character relations of [Ar96].

Status 9.8. Lemma 9.1 and Proposition 9.2 are proved here. Corollary 9.3 and Proposition 9.6 are proved here from the list of §6.3, which depends on the unchecked statements of §4. Lemma 9.5 is sketched. Proposition 9.7 is proved under its three assumptions. Theorem D remains an outline. For κ of order n the endoscopic group is a torus and the result on the side of Route A is [JC26, Prop. 5.1].

Example 9.9 ($n=2$)**.** For $G=SL(2)$ there is one non-trivial unramified κ, of order 2. It corresponds to the elliptic element of R_{χ_0} found in Example 3.7. Since κ has order $n=2$, the endoscopic group is a torus. The singular part is the point mass at χ_0 of §10.2, whose weight $\frac{1}{2}$ is given by [JC26, Prop. 6.4].

10 Examples

This section works out what the results say in four cases. For $PGL(2)$ and $PGL(3)$ there is no singular part, and the density can be compared with the closed forms of Route A. For $SL(2)$ there is an atom. For $SL(3)$ we state a prediction that has not yet been tested.

10.1 $PGL(2)$

This is the simplest case: rank one, with connected center. Only the absolutely continuous part appears.

Here $Z(G)$ is connected, all R-groups are trivial, and Theorem 7.1 has only the M_0 term. Theorem B says that $I_B\rho$ is the smooth function $-\frac{1}{2}J_A(\chi^{-1},\phi)$. By [JC26, Thm. 6.1, Rem. 6.2], for γ

of type (U_r),

$$I(\chi,\gamma)\rho(\chi) = \frac{Q}{2}\cdot\frac{|B_r|(2-2\cos\theta)+q^r(2\cos\theta-2q^{-1})}{1-2q^{-1}\cos\theta+q^{-2}} \qquad (y=\chi_\alpha(\pi)=e^{i\theta}),$$

which is smooth, as Theorem B predicts. Conjecture 8.1 identifies this with a weighted character.

Status 10.1. Next step: compute $J_A(\chi^{-1},\phi)$ directly, for ϕ the normalized characteristic function of γK_1. This is a trace on the $(q+1)$-dimensional space of level-one vectors, which can be done numerically, and the result compared with the display above. This will fix the constant c_G and the sign.

10.2 $SL(2)$

This is the simplest case with a singular part. It comes from the one non-trivial R-group of Example 3.7.

At the unramified quadratic character χ_0, $R_{\chi_0}=\{1,s\}$ and s acts by -1 on $\mathfrak{a}_0$, so s is elliptic. The corresponding virtual character is $\tau=\pi^+-\pi^-$, where $I(\chi_0)=\pi^+\oplus\pi^-$ and π^+ contains the spherical vector. So $\operatorname{tr}\tau(f)=f^\vee(\chi_0)$, and the $M=G$ term of Theorem 7.1 is a point mass at χ_0 with weight $c_\tau\operatorname{tr}\tau^\vee(\phi)$. For ϕ concentrated near an unramified elliptic γ of depth 0, $\operatorname{tr}\tau^\vee$ is the κ-character, and [JC26, Prop. 6.4] gives the weight $\frac{1}{2}$. This fixes c_τ for $SL(2)$, and the result agrees with Route A.

10.3 $PGL(3)$

This is a rank-two case with connected center. Route A has a closed form and numerical results to compare with.

All R-groups are trivial, so Theorem C predicts no singular part, in agreement with the numerical result of [JC26, App. A]. Theorem B predicts that $I\rho$ is smooth, which agrees with the closed form [JC26, (A.4)]:

$$I(\chi,\gamma)\rho(\chi) = \frac{(q-1)^2(q^2+q+1)}{2q^6}\cdot\frac{2(q^2+q+1)-q\sum_{\alpha\in\Phi}z^\alpha}{\prod_{\alpha\in\Phi}(1-q^{-1}z^\alpha)}.$$

Conjecture 8.1 predicts that this is $c_G J_{M_0}(\chi^{-1},\phi)$ for ϕ concentrated near γ.

10.4 $SL(3)$: a prediction

This is a rank-two case with non-trivial R-groups. Nothing is proved here. We state what Theorem C and the outline of §9 predict.

Let $G=SL(3)$, and let γ be regular elliptic in the norm-one torus of the unramified cubic extension. Up to W, the unitary unramified characters with non-trivial R-group and an elliptic element in it are the points

$$\chi_\zeta=(1,\zeta,\zeta^2) \qquad (\zeta^3=1,\ \zeta\neq 1),$$

in coordinates where χ is a triple modulo scalars (as for the dual group $PGL(3,\mathbb{C})$). At χ_ζ a 3-cycle in W fixes χ and acts on $\mathfrak{a}_0$ without fixed vectors. Here $R_{\chi_\zeta}\cong\mathbb{Z}/3$, $I(\chi_\zeta)$ has three constituents,

and there are two elliptic virtual characters, attached to the two non-trivial elements of R. The Satake parameter of χ_ζ is $\mathrm{diag}(1,\zeta,\zeta^2)$, which is also the parameter χ_κ attached to the characters κ of order 3 in [JC26, Prop. 5.1].

Prediction. The spectral measure of O_γ on $SL(3)$ is $I\,d\mu$ plus a point mass at χ_ζ. Its weight is the sum of the two κ-contributions, $\frac{1}{3}(\Delta_\kappa^{-1}+\Delta_{\kappa^2}^{-1})$ in the notation of [JC26, (5.4A), Prop. 5.1], which is $\frac{2}{3}$ if both transfer factors equal 1. The constant c_S cancels, as in (9.1). There are no singular contributions on one-dimensional subtori: those would require an elliptic element of an R-group for $M \cong GL(2)$, and the R-groups of unramified principal series of $GL(2)$ are trivial.

Status 10.2. To be tested numerically on the building of $SL(3)$, by adapting the $PGL(3)$ code of [JC26, App. A] (`examples/pgl3/`) to vertices of one type and to the cubic norm-one torus. The predicted mass $\frac{2}{3}$ depends on normalizations that have not been checked.

11 Summary of results

This section collects the results of the paper, with the status of each as it is stated where the result appears. No status here is meant to be stronger than in the body. The remaining work and the open questions are listed in §12.

Result	Content	Status
Lemma 3.4	Normalized weighted characters of spherical functions vanish for $M \neq L$.	Proved here from (3.3). Arthur's formula uses unnormalized weighted characters (Remark 4.1). See item (N1).
Proposition 5.1	The geometric side reduces to $I(f, \phi)$.	Proved here, from the statement of [Ar91, Thm. 12.2] and its measures.
Proposition 6.1	The spherical factor contributes only $f^{\vee}(\chi)$.	Proved here from (4.5), whose form in the local setting is being checked. See item (N2).
Lemma 6.2	Functions supported on G_{ell} are cuspidal.	Proved here from van Dijk's formula [vD72].
Lemma 6.3	Square-integrable representations of Levi subgroups $L \neq M_0$ have no hyperspecial-fixed vectors.	Proved here from Macdonald's Plancherel theorem [Ma71], in a form that has not been checked. See item (N3).
Theorem 7.1 and Theorems A, B, C	The spectral measure exists. Its absolutely continuous part is a weighted character, and its singular part comes from elliptic virtual characters.	Proof in outline, with the normalizations of §12 assumed. In rank ≥ 2 the formula for the density may need further terms (Remark 4.1, Status 7.2). See items (N1), (N2) and (N4).
Theorem D	For $SL(n)$ the singular part is given by endoscopy.	Outline only. See §9.4 and item (N8).
Lemma 9.1, Proposition 9.2	The R-groups and the elliptic data of $SL(n)$.	Proved here.
Corollary 9.3, Proposition 9.6	For $SL(n)$ the singular terms and the κ-orbital integrals have the same index set and the same supports.	Proved here from the list of §6.3 and from Lemma 9.5, which is sketched.
Proposition 9.7	A summed form of Theorem D.	Proved here under its three assumptions, of which two are not known.
Lemma 8.2, Proposition 8.3	If both routes give a measure, the densities and the singular parts agree.	Proved here. Conditional on the two hypotheses stated.
Lemma 8.4	The weighted character $J_{M_0}(\chi, f)$ is W-invariant in χ.	Proved here, using that it does not depend on the parabolic subgroup.
Corollary 8.5	Conjecture 8.1 for $PGL(2)$.	Proved here, assuming Theorem 7.1 for $PGL(2)$.
Conjecture 8.1	The density I_B equals the T-independent part of the kernel of Route A.	Conjectural. See item (N6).
§10.4	A point mass at χ_ζ for $SL(3)$.	A prediction. It has not been tested numerically, and it depends on normalizations that have not been checked. See item (N7).

The examples of §10 for $PGL(2)$, $SL(2)$ and $PGL(3)$ compare Theorem 7.1 with results of [JC26]. The rank-one computation that would fix the constants c_G and c_τ directly is item (N5).

12 Status and what remains

This section lists the steps that remain before the results of the paper are fully proved. The items are cited by number throughout the paper. They also serve as the list of open questions.

(N1) *Normalizing factors.* Check that (3.2) satisfies Arthur's conditions for normalizing factors and agrees with [Ar98] in the unramified case (non-split groups included). Compute, from (4.4), the terms that relate Arthur's unnormalized weighted characters to the normalized ones in rank ≥ 2 (Remark 4.1), starting with $PGL(3)$, and correct Theorem B and Theorem 7.1 accordingly.

(N2) *Arthur's formula.* The statements of (4.1) and (4.2) now follow [Ar91, Thm. 12.2]. Fix the elliptic form from [Ar93], including the relation between $\Pi_{\mathrm{disc}}(M)$ with its coefficients and $T_{\mathrm{disc}}(M)$, the constants and measures, and the splitting formula (4.5) in the local setting.

(N3) *Lemma 6.3.* Check the form of Macdonald's Plancherel theorem that the proof uses.

(N4) *Theorem 7.1.* Write the proof in full, including the smoothness of $\chi \mapsto J_{M_0}(\chi,\phi)$ in the compact picture.

(N5) *Rank one.* Compute $J_A(\chi,\phi)$ for $PGL(2)$ and $SL(2)$ numerically, and compare with [JC26, Thm. 6.1, Prop. 6.4]. This fixes c_G and c_τ.

(N6) *Conjecture 8.1.* Give a direct proof, at least in rank one.

(N7) $SL(3)$. Run the numerical test of the prediction.

(N8) *Theorem D.* Carry out the three steps listed at the end of §9.4, and prove (9.2) for prime n.

A Glossary

This glossary gives short definitions of the terms used in this paper, with pointers to where they are used. Terms already in the glossary of [JC26] are repeated only briefly.

Unramified group; hyperspecial K
A reductive group over F, quasi-split and split over an unramified extension. K is the stabilizer of a hyperspecial vertex of the building; for split G, $K = G(\mathcal{O})$.

Spherical function Γ_χ; Satake transform $f^\vee$
Γ_χ is the bi-K-invariant matrix coefficient of the spherical vectors of $I_B(\chi)$. The Satake transform is $f^\vee(\chi) = \operatorname{tr} I_B(\chi, f)$ for $f \in \mathscr{H}(G,K)$, a W-invariant Laurent polynomial in χ.

Plancherel measure $d\mu = \rho\, d\chi$
The measure on $\widehat{T}^u/W$ for which $f \mapsto f^\vee$ is an isometry from $L^2(K\backslash G/K)$ (Macdonald). ρ is proportional to $|c(\chi)|^{-2}$.

Principal series $I_P(\chi)$; compact picture
Normalized induction from a character of $M_0(F)$, realized on functions on K so that all $I_P(\chi)$ share one space.

Intertwining operator $J_{P'|P}(\chi)$
The operator $I_P(\chi) \to I_{P'}(\chi)$ given by integration over $N_{P'} \cap N_{\bar{P}}$, meromorphic in χ (§3.2).

Normalizing factor; normalized operator $R_{P'|P}$
A scalar $r_{P'|P}(\chi)$ with $R = r^{-1}J$ unitary and holomorphic on the unitary axis and multiplicative. In the unramified case it is the product of $L(0,\chi_\alpha)/L(1,\chi_\alpha)$ over the relevant roots, and R fixes the spherical vector (3.3).

(G,M)**-family**
A family $\{c_Q(\Lambda)\}_{Q \in \mathscr{P}(M)}$ of smooth functions that agree on common walls. Its limit $c_M = \lim_{\Lambda \to 0} \sum_Q c_Q(\Lambda)\theta_Q(\Lambda)^{-1}$ exists. For a constant family $c_M = 0$ unless $M = G$ [Ar81].

Weighted orbital integral $J_M(\gamma, f)$
An orbital integral against the weight function v_M attached to a (G,M)-family. It is not invariant unless $M = G$. These are the terms of the geometric side of Arthur's trace formulas.

Weighted character $J_M(\pi, f)$
$\operatorname{tr}(\mathscr{R}_M(\pi,P)I_P(\pi,f))$, with $\mathscr{R}_M$ the limit of the (G,M)-family of normalized intertwining operators (§3.3). These are the terms of the spectral side. They depend on the normalization.

Splitting formula
The expression (4.5) of a weighted character of $f_1 \otimes f_2$ through weighted characters of f_1 and of f_2 on Levi subgroups, with coefficients $d_M^G(L_1, L_2)$.

Local trace formula
Arthur's identity between the geometric expression (4.1) and the spectral expression (4.2), for $f \in C_c^\infty(G(F) \times G(F))$. It is deduced from two expansions of the constant term $J(f)$ of a truncated integral of the kernel of $R(f)$ on $L^2(G(F))$ [Ar91].

Truncation $u(x,T)$
The characteristic function of the compact set of $x \in G(F)$ whose Cartan component lies in the convex hull of $W \cdot T$. It is the analogue of the set $G(T)$ of Route A.

Regular elliptic element; G_{ell}
A regular semisimple element whose centralizer is an anisotropic torus (modulo the center).

Cuspidal function
$\phi \in C_c^\infty(G(F))$ whose orbital integrals vanish on non-elliptic regular elements; equivalently, $\operatorname{tr}\sigma(\phi) = 0$ for properly induced σ. Functions supported on G_{ell} are cuspidal (Lemma 6.2).

R**-group** R_χ
W_χ / W_χ°, the stabilizer of χ in W modulo the subgroup generated by the reflections in roots α with $\chi_\alpha = 1$. It controls the reducibility of $I_P(\chi)$ (§3.4).

Elliptic element of an R**-group**
An $r \in R_\chi$ represented by w with no non-zero fixed vector in $\mathfrak{a}_0^G$ (or in $\mathfrak{a}_0^M$, for M).

$T_{\mathrm{disc}}(M)$**,** $T_{\mathrm{ell}}(M)$
Arthur's sets of virtual tempered characters: triples (L, σ, r) with σ square-integrable on a Levi $L \subseteq M$ and r in the R-group of σ in M. $T_{\mathrm{ell}}(M)$ is the subset with r elliptic [Ar93].

Absolutely continuous and singular parts
The decomposition $dI_\phi = I_B\, d\mu + dI_\phi^{\mathrm{sing}}$ of the spectral measure (Theorems B and C).

κ-orbital integral; endoscopic group; transfer factor
See [JC26, §5]. For $SL(n)$, κ is a character of $PGL(n,F)$ trivial on the image of $SL(n,F)$. The elliptic endoscopic groups are the H_m, and transfer factors Δ relate their stable orbital integrals to κ-orbital integrals.

Fundamental lemma
The identity, for the spherical Hecke algebra, between κ-orbital integrals and stable orbital integrals on endoscopic groups. It is a theorem (Ngô [Ng10] and others).

Route A, Route B
Route A is the truncation method of [JC26]. Route B, this paper, specializes Arthur's local trace formula.

Hypothesis (R)
The regularity hypothesis of [JC26, §4.3]: each oscillatory coefficient of the truncated kernel, times ρ, is smooth. It is sufficient for the spectral identity but not necessary (§8).

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This paper began with a suggestion of Claude (Anthropic), an AI model. During the 2026 revision of the first author's paper [Jo92], Claude suggested reproving the results of the 1997 version of that paper from Arthur's later work on the local trace formula [Ar91, Ar93]. The first author decided to pursue the suggestion, directed the work and reviewed the drafts.

Claude took part as a co-author. It wrote the outline (`OUTLINE.md`, in the directory of the source), which the plan of the paper follows. It drafted the text, the examples, the literature review, the summary and the glossary, and it made the style revision of Draft 2 and the mathematical revisions of Drafts 3 and 4. Gemini (Google), an AI model, acted as a referee of Draft 3, at the first author's request. The proof of Lemma 6.3, Lemma 8.4 and several clarifications follow its report. The paper has not been refereed by a person.

Review by AI systems is not an independent certification of the results. The errors that remain are the authors'. Further details on the AI usage, and a list of every change between drafts, can be found in the revision log (`revision_log.md`) that accompanies the source.

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